PlotExtractor

User guide

Make and edit plot grids, cut one original-quality image per plot from Phase One drone photos (from the JPEGs or straight from the IIQ raw files), and make training tiles from the middle crop rows of each plot.

Department of Primary Industries and Regional Development
GOVERNMENT OF WESTERN AUSTRALIA

APPN
Australian Plant Phenomics Network

PlotExtractor 2026-10-24

One original-quality image per plot, and training tiles from the middle rows

Phase One RGB · Metashape project + plot shapefile · DPIRD / APPN



raw plot images (original pixels)



middle-3-row tiles, plot 3003

PlotExtractor 2026-10-27 · DPIRD / APPN · 28 October 2026

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1. What PlotExtractor does

PlotExtractor takes a drone flight that has been processed in Agisoft Metashape and gives you a plot grid, one image per plot and training tiles. The tabs are numbered in the order you normally use them:

Tab	What it does
1 Generate plot grid	Draws the plot shapefile on an RGB orthomosaic from the 4 trial corners and the layout (no Metashape needed).
2 Edit plot grid	Moves, reshapes, grows/shrinks, widens/lengthens or shifts plots on the RGB orthomosaic; corrects the grid to the drilled rows.
3 IIQ conversion	Converts Phase One IIQ raw files to TIFF, GeoTIFF (georeferenced) or JPEG.
4 Extract plots	Our method: one image per plot cut from the Phase One JPEGs (pixels identical to the photos), or from the orthomosaic.
5 IIQ plot extraction	The same extraction, with each plot developed straight from its IIQ raw file (approach of APPN, The University of Queensland, with our changes).
6 Training tiles	Square tiles from the middle crop rows of every plot, kept away from the bare plot ends.
7 QA	Checks the plot images against their source, the tiles against the plot images, and the rows; pictures per plot.
8 Acknowledgements	Credits APPN (The University of Queensland) and Phase One for the IIQ plot extraction, and what we changed.

The current grid. The grid saved in tab 1 or tab 2 becomes the current grid: tab 2 opens it for editing, and tabs 4 and 5 use it by default. Each of tabs 4 and 5 can still use any other plots file (Browse).

Coordinate system. All outputs are written in **EPSG:7850 (GDA2020 / MGA zone 50)** by default: the grids of tabs 1 and 2 (reprojected from the orthomosaic's or input file's CRS) and the georeferenced images of tabs 3-5. Each of these tabs has an Output CRS / Save in CRS field for another EPSG code (e.g. 7851 for zone 51). Inputs may be in any CRS. Loading an orthomosaic in tab 1 or 2 sets all these fields to the orthomosaic's CRS (a lon/lat orthomosaic gives its local MGA2020 zone, 7850 at York); you can still change them.

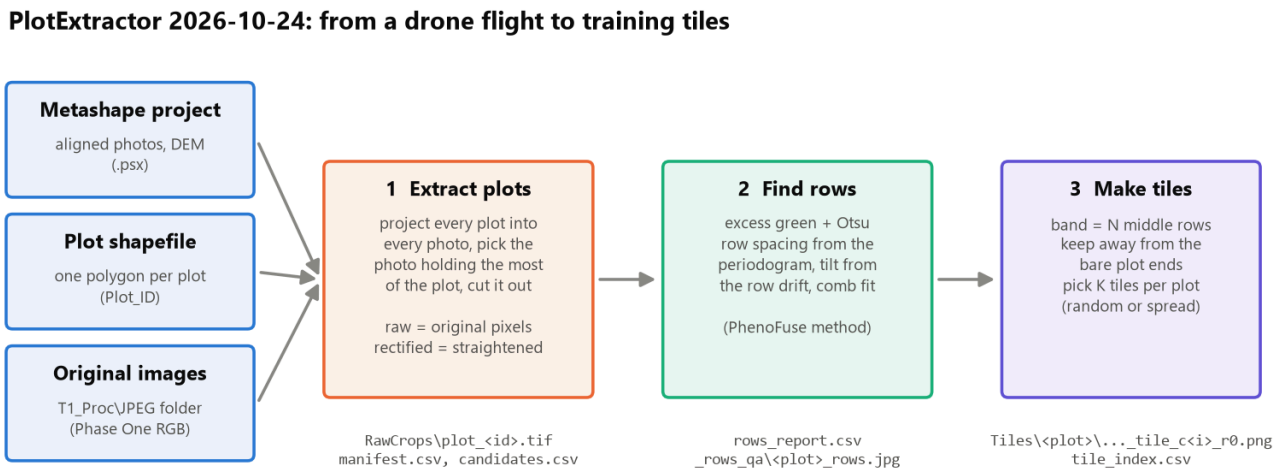


Figure 1. *The core steps (grid, plot images, tiles) and the files each step writes.*

It was built for Phase One RGB cameras (the T1_Proc\JPEG images) but works with any camera that Metashape has aligned. With the recommended settings the plot images are **pixel-for-pixel identical to the original photos** (chapter 10 and the technical report).

2. Quick start: the recommended way

For Phase One RGB images and original-quality plot images plus middle-row training tiles:

1. Start PlotExtractor\PlotExtractor.exe.
2. **Tab 1 or 2:** make the plot grid on the RGB orthomosaic (tab 1), or load your grid in tab 2 and fine-tune it; **Save & use** makes it the current grid.
3. **Tab 4 (Extract plots):** choose the Metashape project (.psx) and, under Raw images folder, the flight's original T1_Proc\JPEG folder; the plot grid is already filled in. Choose an output folder. (For the highest image quality use tab 5 instead, chapter 8.)
4. Set **Margin 0.10 m**, **Format GeoTIFF**, tick **Raw image crop per plot** with style **Masked to plot**, and leave **Convert raw crops to sRGB** off.
5. Set **Limit 3** and press **Run extraction** as a test. Open a few crops in RawCrops\. If they look right, set Limit back to 0 and run all plots.
6. **Tab 6 (Training tiles):** Plot crops folder = <output>\RawCrops; choose a tiles output folder.
7. Tick **Tile only the middle 3 rows**, set **sown rows per plot** (5 at York), **Tiles per plot** (e.g. 4), pick **random** or **spread**, keep tiles **0.5 m** from the plot ends.
8. Format **tif**, Bits **same as plot image**, TIFF compression **LZW**, tick **Georeference tiles**, **Link tiles to the source photo / IIQ**, **One folder per plot** and **QA pictures**, press **Make tiles**. Every tile is then an exact copy of its window in the plot image.
9. Check _rows_qa\ (one picture per plot) and rows_report.csv, and run **tab 7 (QA)** on the output.

Recommended settings for Phase One RGB images

Extract plots	
Raw images folder	**the original T1_Proc\JPEG folder of the flight**
Format	**GeoTIFF (geotagged, lossless)**
Raw crop style	**Masked to plot (100% original pixels)**
Convert to sRGB	**off (keep the captured colours)**
Margin around plot	**0.10 m (outer rows whole when the grid is a little off)**
Limit	**3-5 for a test run, then 0 = all plots**
Training tiles	
Plot crops folder	**<output>\RawCrops**
Tile only the middle N rows	**on, N = 3**
Sown rows per plot	**the trial's sown rows (5 at York); 0 = detect**
Square tiles spanning the band	**on**
Tiles per plot	**e.g. 4 (0 = all that fit) · pick: random or spread**
Keep tiles … m from plot ends	**0.5 m**
Format / folders	**PNG · one folder per plot · QA pictures on**

Figure 2. *Recommended settings at a glance.*

Video: docs\PlotExtractor_Demo.mp4 walks through these steps in the real application (about 7 minutes, narrated; recorded with the earlier tab layout, before the tabs were numbered).

3. Installing and starting

- **Application:** download the release zip from github.com/appndpird/plot-extractor (Releases), unzip it and run PlotExtractor\PlotExtractor.exe. Keep the whole folder together. A console window opens next to the main window and shows progress and errors.
- **From source:** PlotExtractor.bat or python plot_extractor.py with the Python 3.11 that has the Metashape 2.3.1 module, rasterio, Pillow and pyshp.
- **Licence:** tabs 3 (GeoTIFF only), 4 and 5 need a Metashape licence (the floating licence server; VPN if off-site). Tabs 1, 2, 6 and 7 do not.
- **Phase One SDK:** tabs 3 and 5 develop IIQ files with the Phase One ImageSDK. The DPIRD build carries it; otherwise point **Phase One SDK folder** at an SDK obtained from Phase One. The status line under it must say OK.
- **Settings** are remembered in config.json next to the application.

4. Tab 1 · Generate plot grid

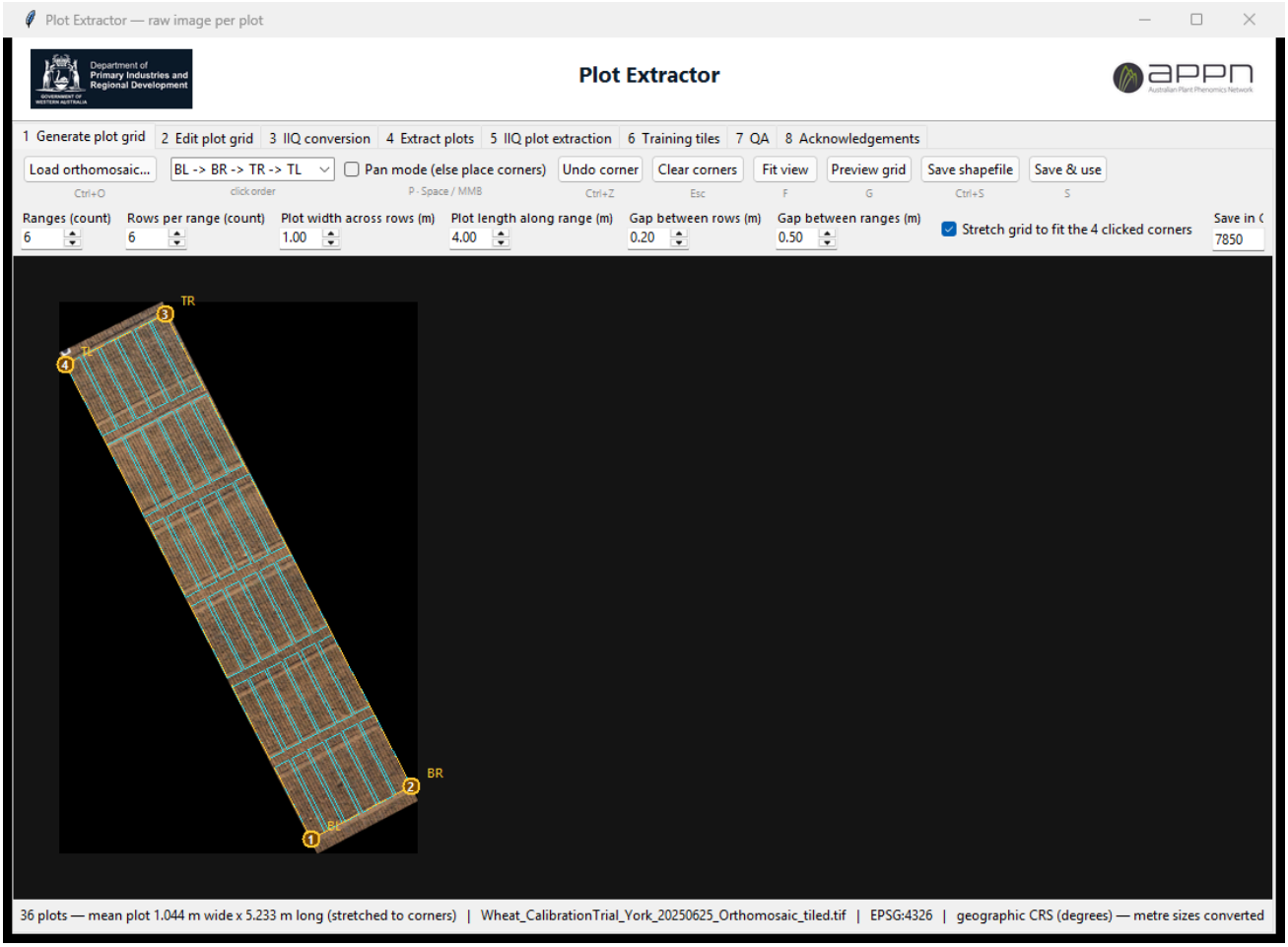


Figure 3. *Tab 1: the 4 trial corners clicked on the RGB orthomosaic and a 6 x 6 grid previewed (York 2025).*

Use it when you do not have a plot shapefile. It needs only an RGB orthomosaic GeoTIFF, no Metashape licence.

1. **Load orthomosaic…** (Ctrl+O). Mouse wheel zooms, Space or middle-drag pans, F fits the view.
2. Click the **4 corners of the trial** in the order shown in the drop-down. Right-click undoes the last corner.
3. Enter **Ranges**, **Rows per range**, plot width and length and the gaps. Keep **Stretch grid to fit the 4 clicked corners** on for rotated trials.
4. **Preview grid** (G) and check the reported plot size, then **Save & use** (S): the shapefile is written in the **Save in CRS** (default EPSG:7850), becomes the current grid and opens in tab 2.

Fine-tune the grid in **tab 2** (chapter 5).

5. Tab 2 · Edit plot grid

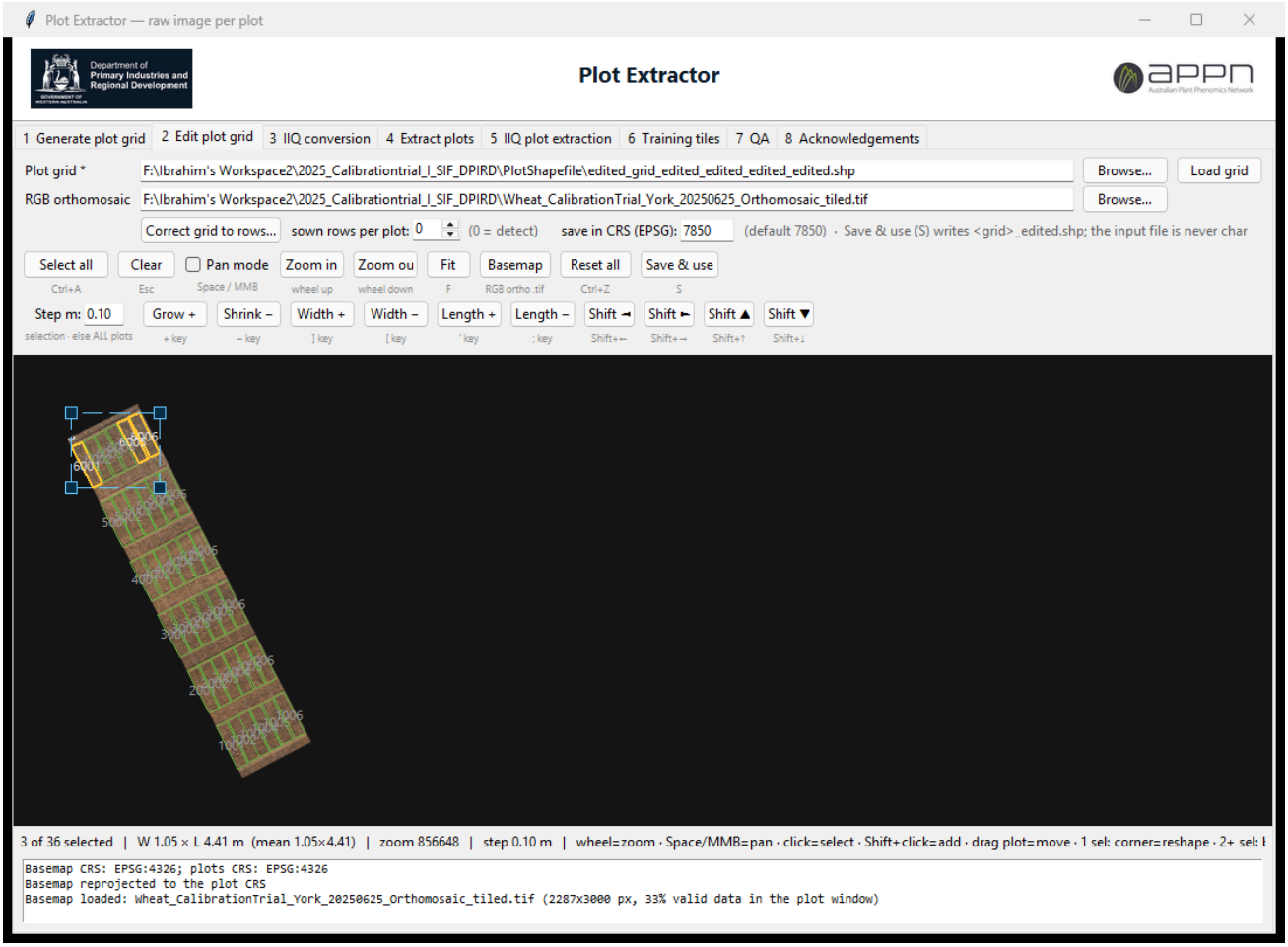


Figure 4. *Tab 2: the York grid on the RGB orthomosaic, three plots selected (blue handles resize them together).*

Edits a plot grid on the RGB orthomosaic. The file you load is never changed: Save & use writes <grid>_edited.shp next to it and makes that the current grid for tabs 4 and 5.

1. **Plot grid:** filled with the current grid (from tab 1, or the last grid you saved); Browse for another and press **Load grid**.
2. **RGB orthomosaic:** the trial's orthomosaic GeoTIFF; it is drawn under the plots (any CRS, reprojected to the plots). **Basemap** on the toolbar loads another one.
3. **Select:** click a plot, Shift/Ctrl-click to add, drag a box around plots, Ctrl+A for all. With nothing selected the tools act on **all** plots.
4. **Edit:** drag plots to move them; with one plot selected drag its corners; with 2+ selected drag a blue handle to resize them together. The buttons change the selection by the **Step** (m): Grow/Shrink, Width ±, Length ±, Shift ◄ ► ▲ ▼ (keys + –] [' ; and Shift+arrows).
5. **Save & use** (S). The edited grid is written in the **save in CRS** (default EPSG:7850). Reset all (Ctrl+Z) goes back to the loaded grid.

Correct grid to rows… measures every plot against the orthomosaic and centres it on its drilled rows (set the sown rows per plot, or 0 to detect them). It writes <grid>_refit.shp with an audit CSV and asks whether to use it. It refuses when the rows run continuously across plots, because then the imagery has no plot edge to fit.

The editor is for RGB orthomosaics; the bottom box shows its messages.

6. Tab 3 · IIQ conversion

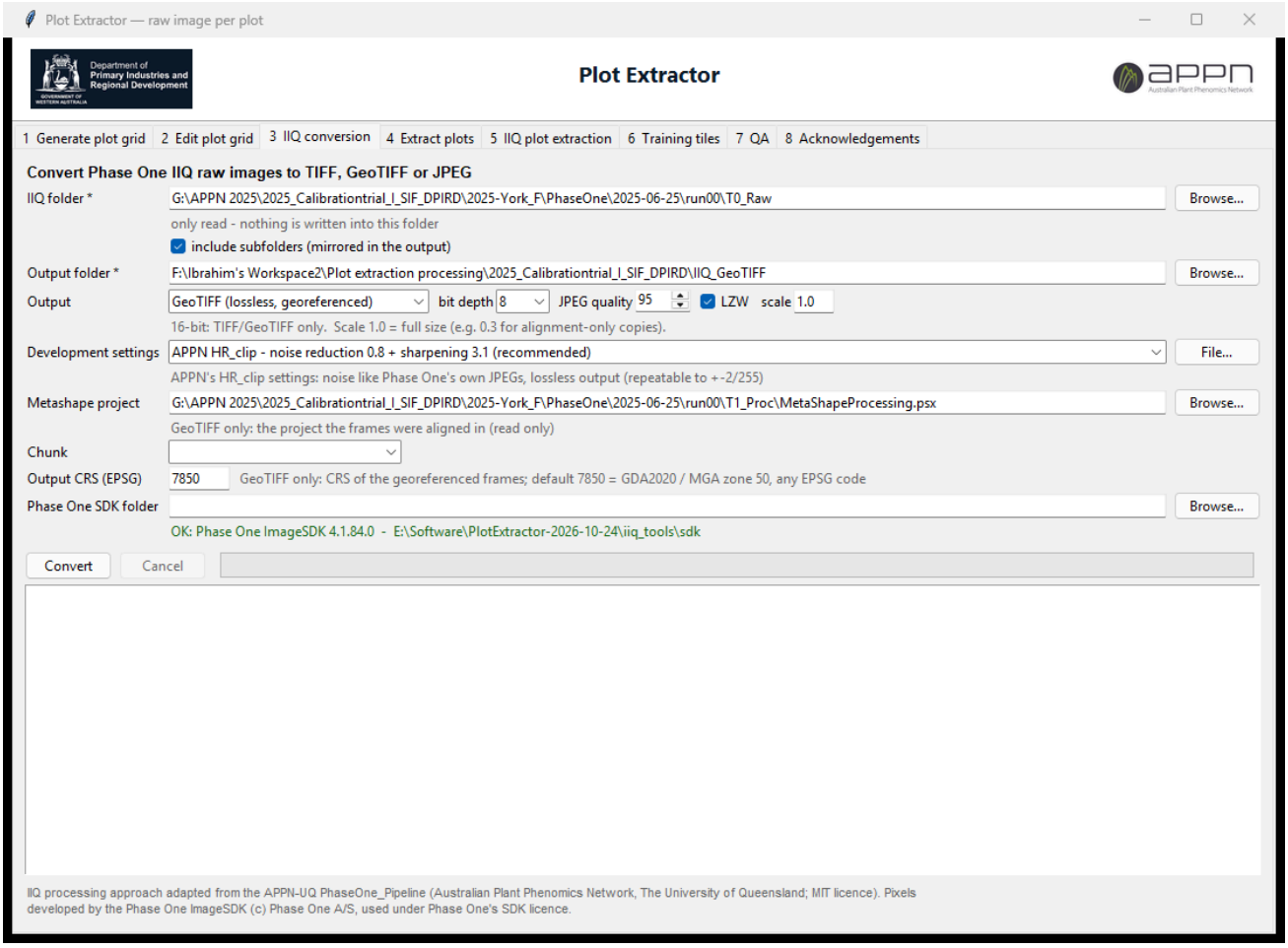


Figure 5. *Tab 3: all IIQ files of a flight to georeferenced GeoTIFFs with the APPN HR settings.*

Converts Phase One IIQ raw files to images with the Phase One ImageSDK. The IIQ folder is only read; an output folder inside it is refused.

Field	What it is	Recommended
IIQ folder	The flight's T0_Raw folder (subfolders optional, mirrored in the output)	T0_Raw
Output	TIFF (8 or 16 bit, LZW), GeoTIFF (georeferenced) or JPEG; scale 1.0 = full size	GeoTIFF, 8 bit, 1.0
Development settings	APPN HR_clip (noise reduction 0.8, sharpening 3.1), Neutral (no processing), APPN template, or a Phase One settings file	APPN HR_clip
Metashape project / Chunk	GeoTIFF only: the project the photos were aligned in. Each frame gets ground control points from its aligned camera and the DEM and an affine georeference (a lon/lat project is written in the local UTM/MGA zone)	The flight's .psx
Output CRS (EPSG)	GeoTIFF only: CRS of the georeferenced frames	7850 (default)
Phase One SDK folder	Blank = the SDK shipped with the app	Blank

Each output keeps the camera EXIF/GPS. conversion_log.csv lists every file, and IIQ_settings_used.json the settings. Checked on York 2025: a converted frame lies within 1 cm of the orthomosaic.

7. Tab 4 · Extract plots

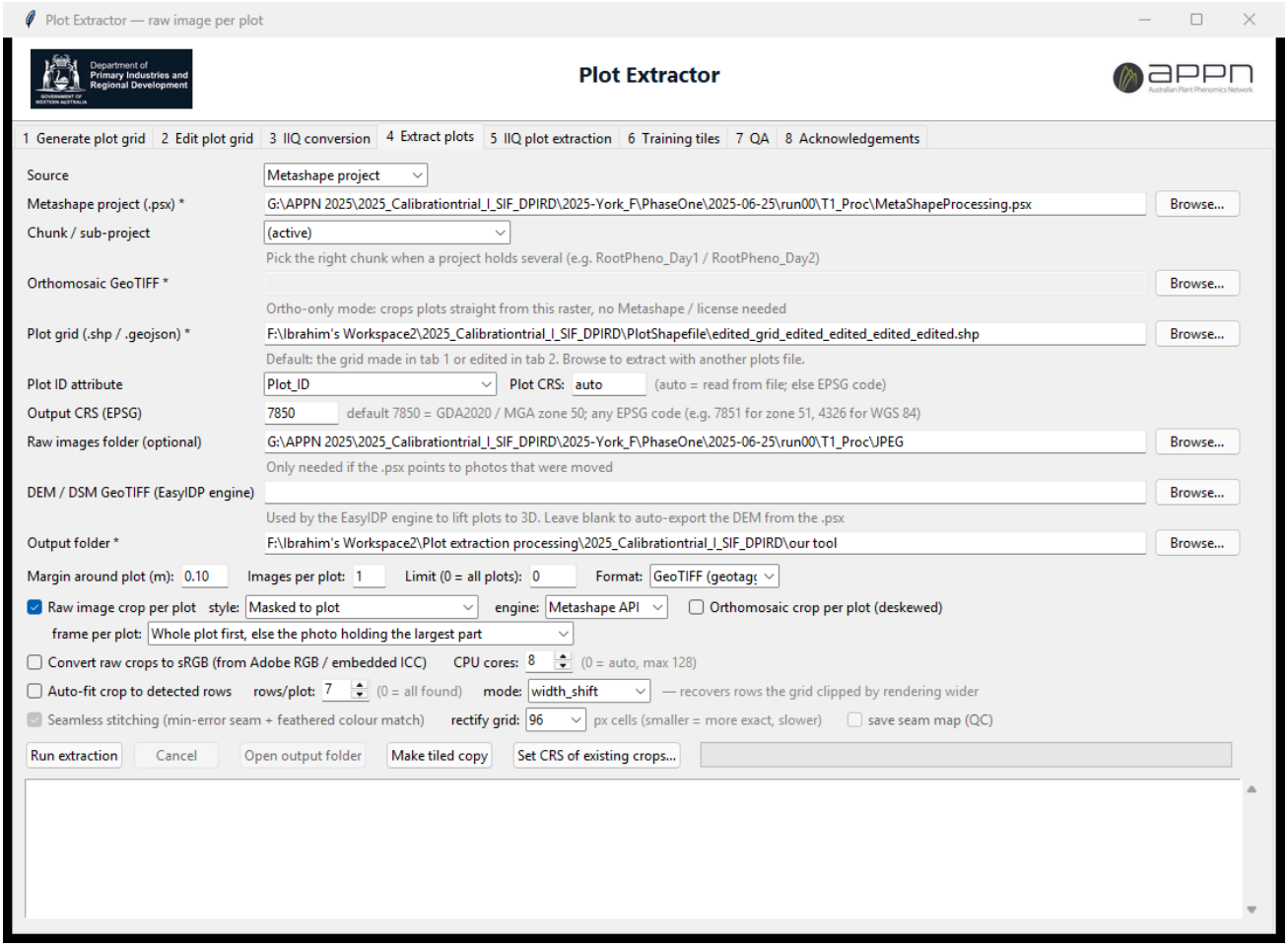


Figure 6. *Tab 4, Extract plots, with the recommended settings for the York calibration trial.*

7.1 The fields

Field	What it is	Recommended
Source	Metashape project (raw + ortho crops) or Orthomosaic GeoTIFF (ortho crops only, no licence)	Metashape project
Metashape project (.psx)	The flight, aligned and georeferenced, with a DEM	The run's .psx in T1_Proc
Chunk / sub-project	When a project holds several chunks	The chunk of this flight
Plot grid	Shapefile or GeoJSON, one polygon per plot, any CRS. Filled with the current grid from tab 1/2; Browse for another file	The current grid
Plot ID attribute	Field used to name the files (plot_<id>.tif)	Plot_ID (picked automatically)
Output CRS (EPSG)	CRS of the GeoTIFF georeference of every crop	7850 (default)

Field	What it is	Recommended
Raw images folder	Where the original photos are. Preferred over the paths stored in the project	T1_Proc\JPEG of the flight
Output folder	Gets RawCrops\, manifest.csv, candidates.csv	A new folder per flight
Margin around plot (m)	Extra ground on every side of the plot	0.10 (0.2 if the grid is far off)
Images per plot	Extra views per plot (plot_<id>_2 ...)	1
Limit	Only the first N plots, for a test run	3-5 to test, then 0 = all
Format	GeoTIFF (lossless, geotagged), PNG, TIFF, JPEG (lossy)	GeoTIFF
Raw image crop per plot · style	How each plot is cut (section 7.2)	On · Masked to plot
Orthomosaic crop per plot	A straightened orthomosaic crop as well	Optional
Convert raw crops to sRGB	Converts the Adobe RGB colours of Phase One JPEGs to sRGB	Off for original pixels
CPU cores	Parallel image work; each core holds a full frame (~370 MB)	0 (auto) or 8
Frame per plot	Whole plot first, else the photo holding the largest part (or the older nadir-tier order)	Whole plot first
Auto-fit crop to detected rows	Snaps the crop to the detected rows. Needs an upright crop: with Native, Masked or Bounding box it switches the style to Rectified (resampled, may stitch photos). With rows/plot 0 it keeps every row it finds, including the outer rows of the neighbouring plots	Off. If used: rows/plot = the sown rows (e.g. 5)
Seamless stitching / rectify grid	Only for rectified crops	Defaults

Plot only, as drawn in the shapefile: Masked to plot (or Native), margin 0, Auto-fit off, sRGB off. The crop then follows the plot grid exactly, so the grid decides what is in the image: two grids of the same trial (e.g. a refit grid) give crops shifted by the same distance.

7.2 Raw crop styles: which one to use

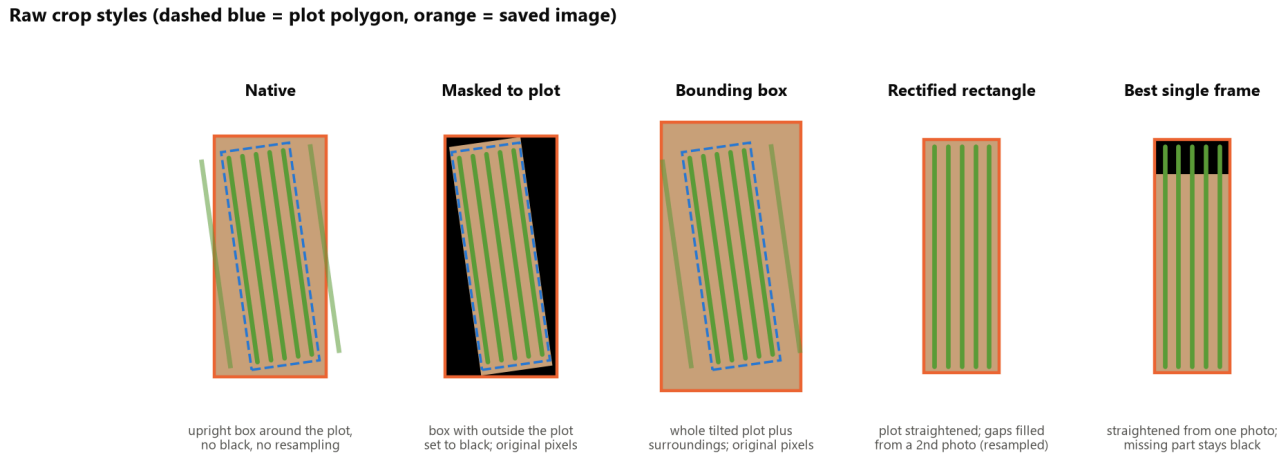


Figure 7. *The raw crop styles on a plot that is slightly tilted in the photo.*

Style	Pixels	Use it when
Masked to plot	Original, unchanged. Outside plot + margin set to black	**Recommended.** You need 100% original quality (training data, counting).
Native (no resample, no black)	Original, unchanged. Upright box, no black	You prefer no black corners and accept some neighbouring ground.
Bounding box	Original, unchanged. Whole tilted plot + surroundings	You want extra context around the plot.
Auto-straighten tilted	Original if the plot is upright, resampled if tilted > 3°	A mixed set where most plots are upright.
Rectified rectangle	Resampled (~5% lower resolution). Gaps filled from a 2nd photo	Plots must be upright and whole, e.g. plot-level measurements.
Best single frame	Resampled; what one photo misses stays black	No stitching allowed at all.

Why not always rectified? Straightening a plot means redrawing every pixel. The result looks the same but is not the original image (about 1.05 original pixels per output pixel). Masked to plot copies the pixels from the photo unchanged.

7.3 What you get

- RawCrops\plot_<id>.tif: one image per plot (GeoTIFF with ground control points, the photo's EXIF and colour profile).
- manifest.csv: one row per plot: source photo, crop size, raw_covered_pct (how much of the plot the image holds), frames_used, raw_style and the pixel box in the photo (src_x0 … src_y1).
- candidates.csv: every photo that sees each plot, with its coverage, so you can see the alternatives.



Figure 8. *A raw (masked) and a rectified crop of the same plot, and the raw crop against the original photo: identical pixels.*

7.4 Check the coverage

Sort manifest.csv by raw_covered_pct. Below 100 means no single photo holds the whole plot, so the raw image is cut at the photo edge. At York this happened to 4 of 36 plots (0.4-3.9% missing). For those plots use the rectified image if you need the whole plot, or see section 12.4 for the flight-planning fix.

8. Tab 5 · IIQ plot extraction

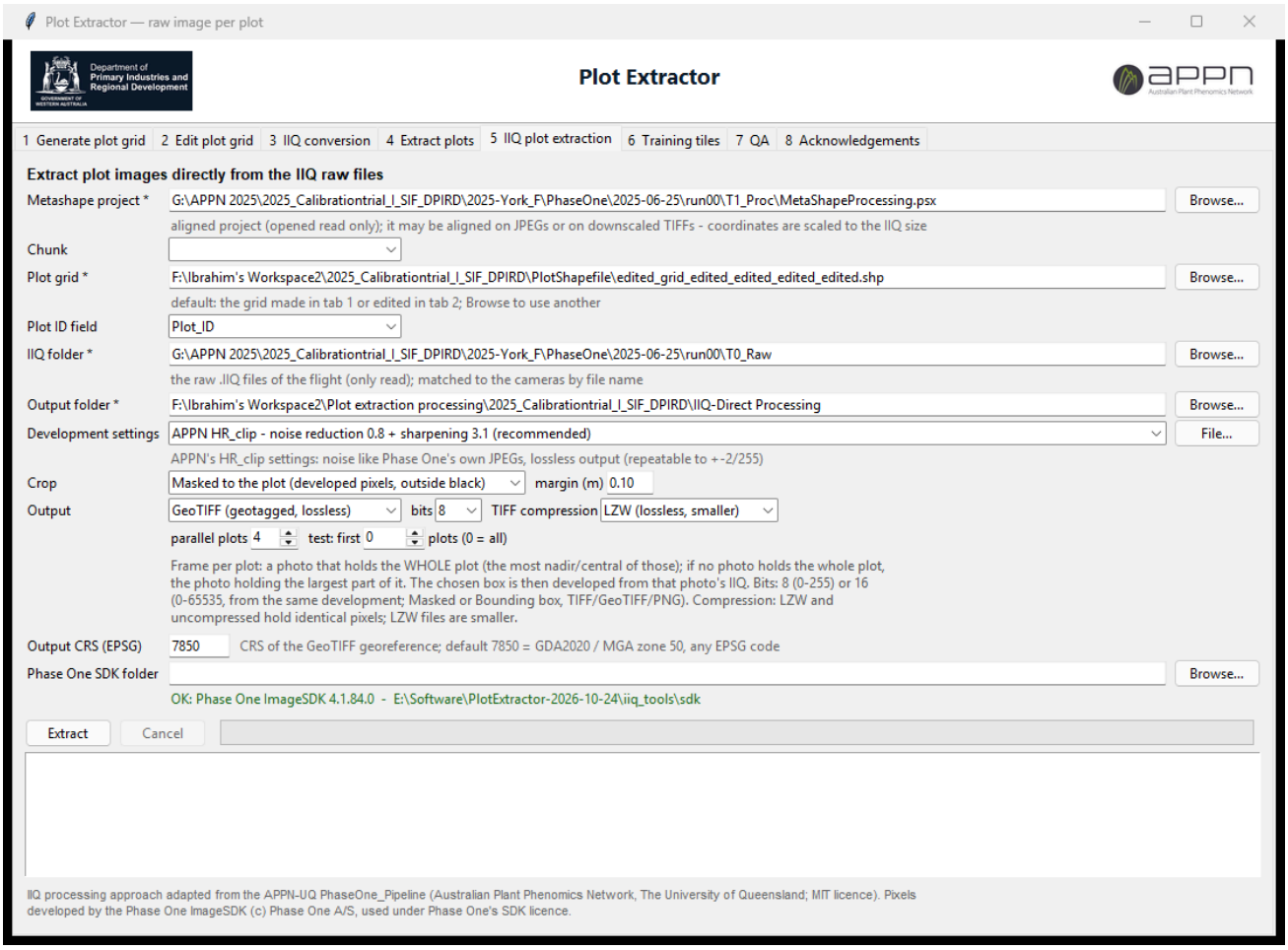


Figure 9. *Tab 5: the York plots developed from the T0_Raw IIQ files with the APPN HR settings.*

The same extraction as tab 4, but each plot's box is developed straight from its IIQ raw file instead of cut from the JPEG. It gives the highest image quality (less noise, more fine detail, no JPEG compression).

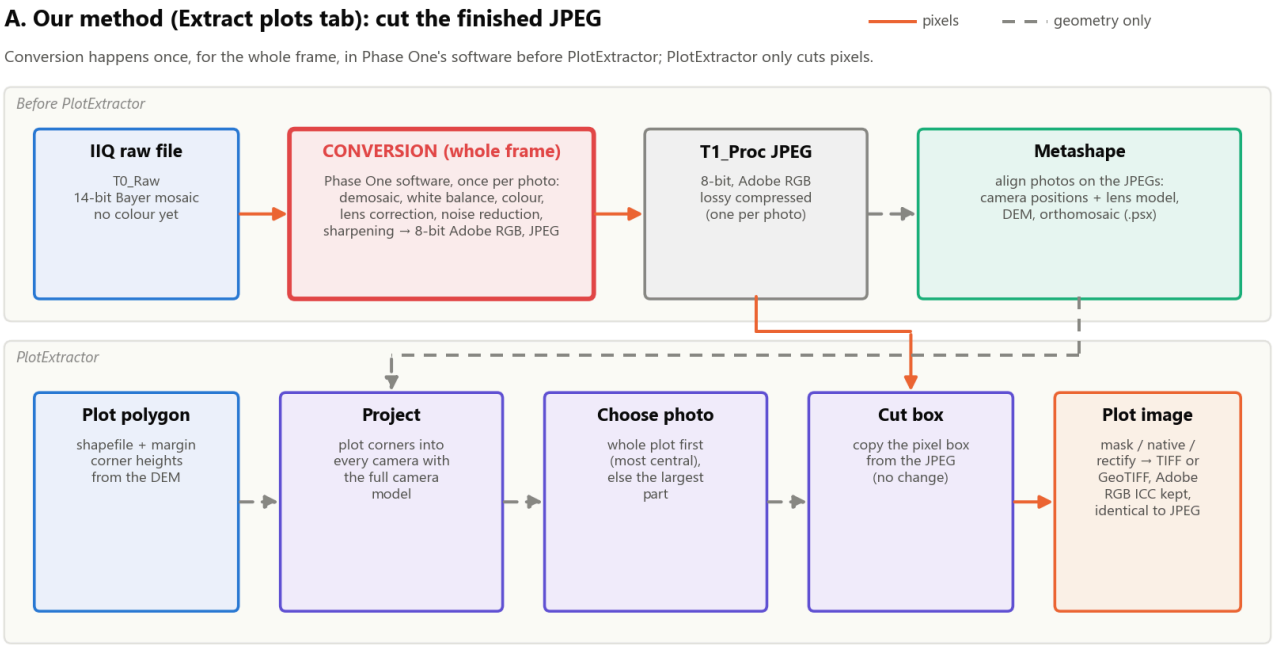


Figure 10. *Tab 4: the conversion is done once, before PlotExtractor, by Phase One's software; the plot is cut from the JPEG unchanged.*

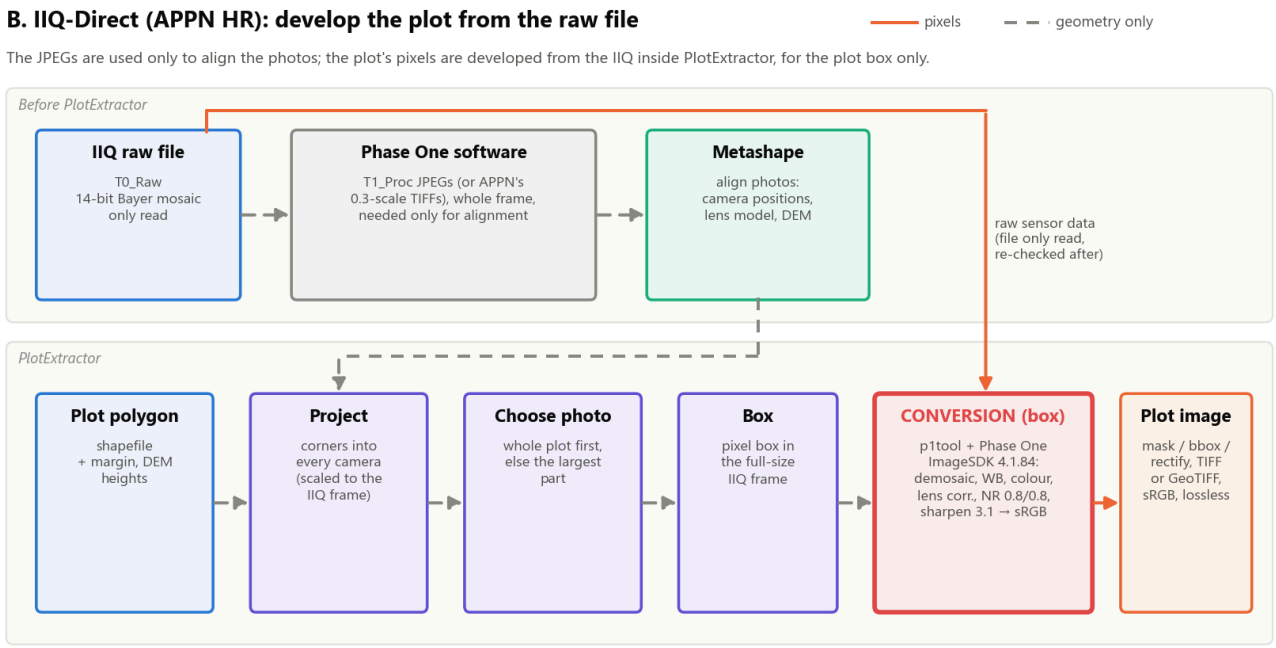


Figure 11. *Tab 5: the JPEGs only give the geometry; the plot box is developed from the IIQ by the Phase One ImageSDK.*

1. **Metashape project:** the aligned flight (aligned on the JPEGs or on downscaled TIFFs; coordinates are scaled to the IIQ size).
2. **Plot grid:** filled with the current grid; Browse for another file. Plot ID field: Plot_ID.
3. **IIQ folder:** the flight's T0_Raw folder (only read; matched to the cameras by file name). Output folder: a new folder.
4. **Development settings:** APPN HR_clip for the best-looking images; Neutral when the least-processed pixels matter (e.g. spectral measurements; noisier). Crop: Masked to the plot, margin 0.10 m.
5. **Output:** GeoTIFF, **bits 8**, **TIFF compression LZW** (section 8.1).
6. Test with **first 3 plots**, check the crops, then 0 = all and **Extract**.

	Tab 4 (JPEG)	Tab 5 (IIQ, APPN HR)
Pixels	Identical to the T1_Proc JPEG	Developed from the raw sensor data
Quality (York 2025, 29 plots)	noise 3.55, detail/noise 5.4	noise 3.13, detail/noise 7.4
Colour	Phase One's JPEG colours (Adobe RGB)	SDK colours (sRGB), a little less green
Speed / needs	Fast; Metashape licence	Slower; also the IIQ files and the Phase One SDK
Repeatable	Bit for bit	Within 1-2/255 (noise reduction); Neutral bit for bit

Use one source per trial and season. Vegetation indices from tab 4 and tab 5 images are not interchangeable (York: green cover 3.0% vs 2.3%, same plot ranking).

8.1 Bits and compression

Option	What it does	Recommended
bits 8	0-255 per channel, as the JPEGs and most training tools expect	Yes
bits 16	0-65535 per channel from the same development (divided by 257 it matches the 8-bit image to 1/255). Masked or Bounding box style; GeoTIFF, TIFF or PNG (not JPEG, not Rectified). Twice the file size; PIL reads 16-bit RGB TIFFs wrongly - use GDAL/rasterio or tifffile	Only for radiometric or exposure work
TIFF compression LZW	Lossless; smaller files (about 10% for camera images)	Yes
TIFF compression None	Uncompressed; the same pixel values as LZW	Only if a tool cannot read LZW

All TIFF and GeoTIFF files are written with GDAL; the developed values are never changed after the SDK (masking only sets the pixels outside the plot to 0). QA (tab 7) re-develops every crop at its own bit depth.

The frame for each plot is a photo holding the whole plot (most central of those) or, if none does, the one holding the largest part. The manifest records pixel_source = iiq; IIQ_settings_used.json and IIQ_run.json record the settings and the SDK; every IIQ used is re-checked unchanged at the end.

8.2 How a plot is made from the IIQ, step by step

A JPEG in T1_Proc\JPEG has already been developed once by Phase One's software, for the whole photo: 8 bit, Adobe RGB, lossy compression. The IIQ in T0_Raw holds the raw 14-bit sensor data from before that. Tab 5 develops each plot from the IIQ itself; the JPEGs are only used by Metashape to align the photos (geometry, no pixels). The example is York calibration trial plot 6006 (25 June 2025).

Step 1 · Find the plot in every photo

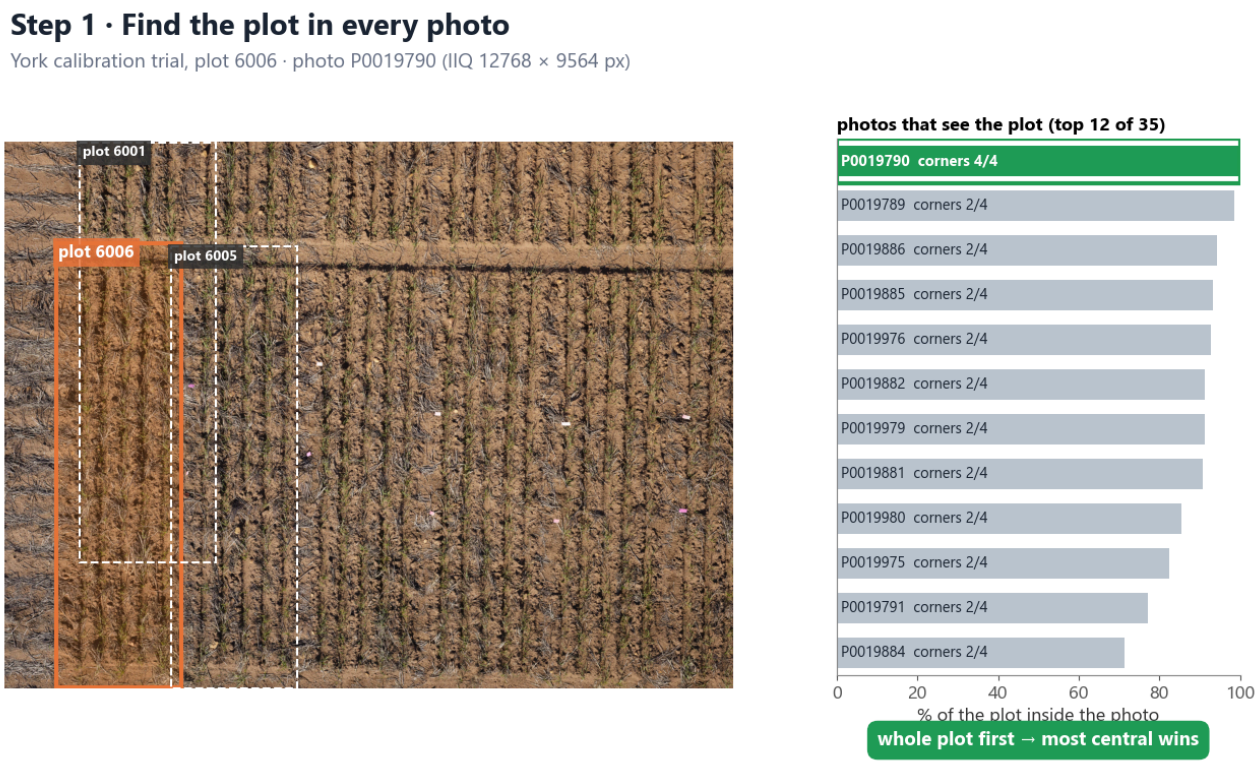


Figure 12. *Plot 6006 in photo P0019790 (left) and the photos that see it (right). Only P0019790 holds all four corners, so it is used.*

The plot polygon plus the margin is placed on the DEM and projected into every photo with Metashape's camera model, lens distortion included (coordinates are scaled to the IIQ size when the project was aligned on smaller images). For plot 6006, 35 photos see the plot. **Photos holding the whole plot come first, and among those the most central one wins** (it looks most straight down). Only when no photo holds the whole plot is the photo holding the largest part used. The candidates are listed in candidates.csv.

Step 2 · Develop only the plot's box

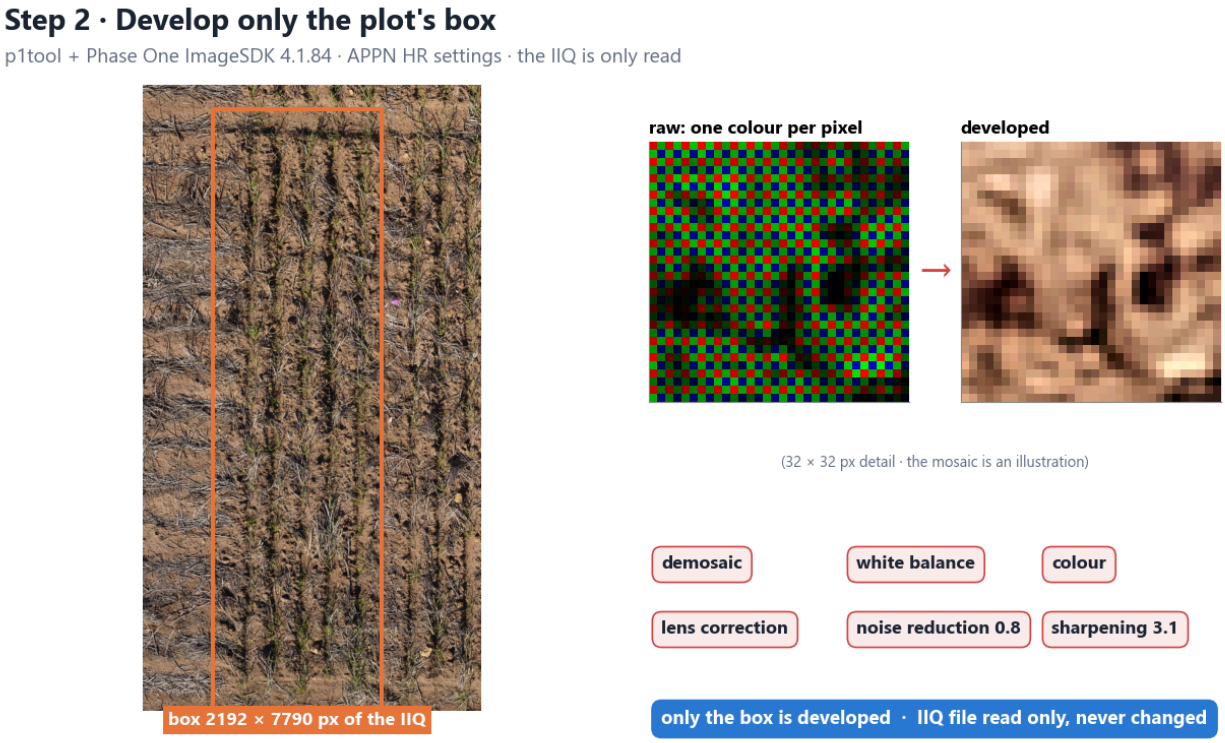


Figure 13. *The plot's box in the chosen IIQ (left) is developed by the Phase One ImageSDK. Right: a 32 x 32 px detail as the sensor saw it (one colour per pixel, an illustration) and developed.*

The plot's pixel box in that photo is passed to p1tool.exe, which uses the Phase One ImageSDK to develop only that box: rebuild full colour from the sensor's colour mosaic (demosaic: each sensor pixel saw only red, green or blue), then white balance, colour, lens correction, noise reduction (0.8) and sharpening (3.1) with the APPN HR settings, at 8 or 16 bit. Only the box is developed, so it is quick; the IIQ file is only read, never changed.

Step 3 · Mask, georeference, save



Figure 14. *The masked plot (left) and the same GeoTIFF in map coordinates, EPSG:7850 (middle).*

Pixels outside the plot polygon are set to black; no other pixel is changed. The image is written with GDAL as a GeoTIFF in GDA2020 / MGA zone 50 (EPSG:7850), with an affine georeference fitted through the plot's ground control points, so it opens in the right place in QGIS. Its tags name the IIQ (SOURCE_IMAGE) and manifest.csv records the photo and the pixel box. 8 or 16 bit and LZW or no compression: section 8.1.

9. Tab 6 · Training tiles

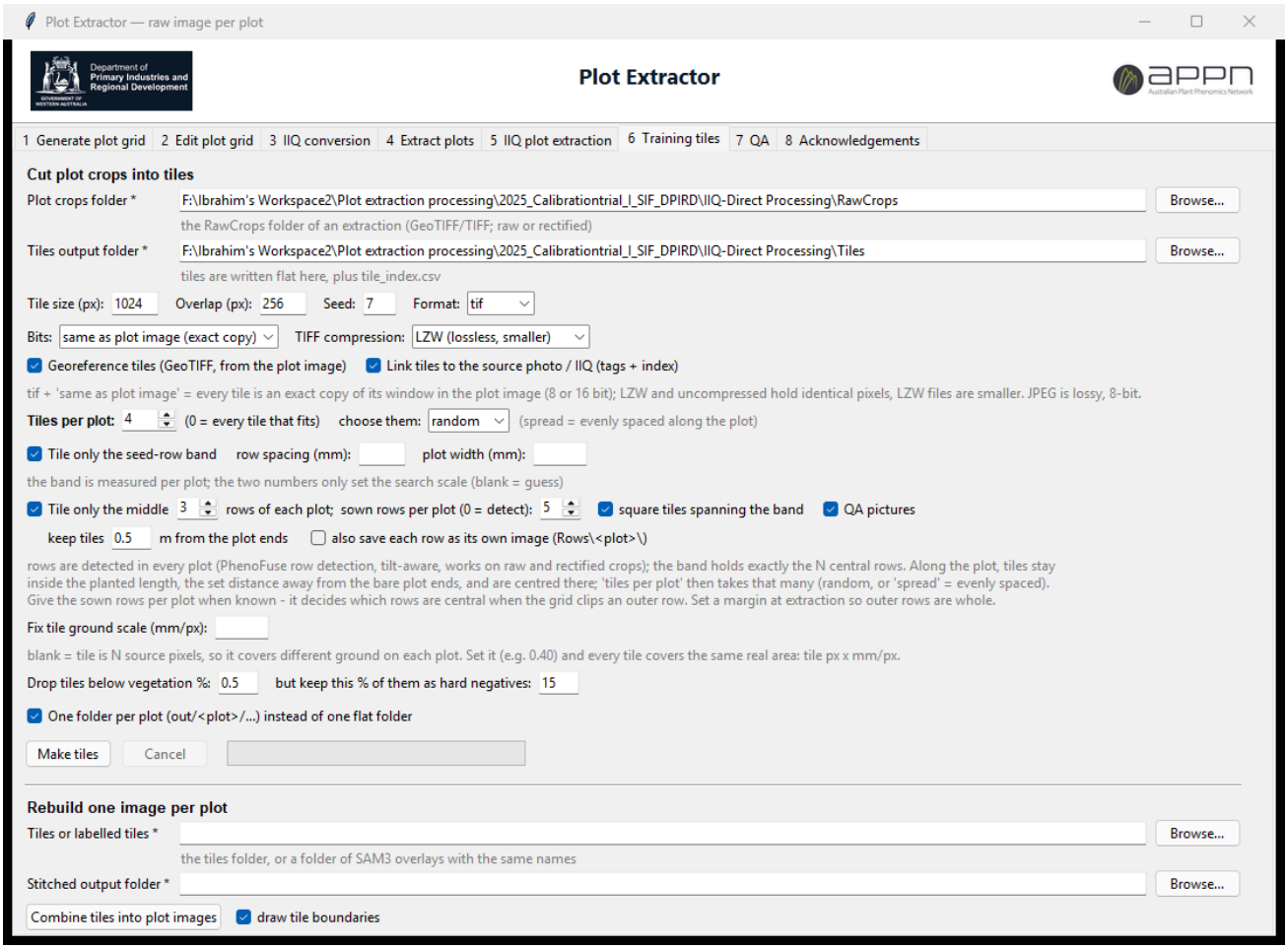


Figure 15. *The Training tiles tab set up for middle-3-row tiles, 4 per plot, 0.5 m away from the plot ends.*

9.1 Middle-row tiles, step by step

1. **Plot crops folder:** the RawCrops folder of an extraction, raw or rectified. Raw crops give original-pixel tiles.
2. **Tiles output folder:** a new folder; tiles plus tile_index.csv are written there.
3. Tick **Tile only the middle N rows** and set N (3).
4. Set **sown rows per plot** when you know it (5 at York, 7 in some trials). 0 = use the rows that are detected.
5. Keep **square tiles spanning the band** on: each tile is exactly as wide as the N rows.
6. Set **Tiles per plot** on its own line under Tile size (0 = every tile that fits), **choose them** random or spread, and **keep tiles … m from the plot ends** (0.5) on the middle-rows line.
7. Format **tif**, Bits **same as plot image (exact copy)**, TIFF compression **LZW**, tick **Georeference tiles** and **Link tiles to the source photo / IIQ**, then **One folder per plot**, **QA pictures** and **Make tiles**.

9.2 The options

Option	What it does	Recommended
Tile only the middle N rows	Tiles only the band holding the N central crop rows of each plot	On, N = 3

Option	What it does	Recommended
sown rows per plot	How many rows were sown. Decides which rows are central when the grid clips an outer row	The trial's value (5 at York)
square tiles spanning the band	Tile side = band width (native pixels). Off = fixed Tile size tiles on the band	On
QA pictures	_rows_qa\<plot>_rows.jpg with rows, band, limits and tiles	On
Tiles per plot	How many tiles each plot gives, in every mode (grid, row band, middle rows). 0 = every tile that fits. If fewer fit, all are written and the log says so	As needed (e.g. 4)
choose them	random = random sample (Seed makes it repeatable); spread = evenly spaced along the plot	random for training, spread for even coverage
keep tiles … m from the plot ends	Tiles stay this far inside the plot ends (after trimming to the planted length)	0.5 m
Format	tif (lossless, can be georeferenced), png (lossless) or jpeg (lossy, 8-bit)	tif
Bits	same as plot image = the plot image's own values (exact copy, 8 or 16 bit); 8 or 16 converts exactly (16 to 8: value/257 rounded; 8 to 16: value x 257)	same as plot image
TIFF compression	LZW (lossless, smaller) or None (uncompressed); identical pixels	LZW
Georeference tiles	TIFF tiles become GeoTIFFs: the plot image's CRS and transform moved to the tile, so a tile pixel has exactly the coordinates of the same pixel in the plot image	On
Link tiles to the source photo / IIQ	Tags in each tile and columns in tile_index.csv: source photo or IIQ, the tile's pixel box in that photo (src_photo_x0, src_photo_y0, w, h), the IIQ settings file, the bit depth	On
One folder per plot	out\<plot>\… instead of one flat folder	On
Tile size, Overlap	Only for the plain grid mode (middle rows off)	-
Fix tile ground scale (mm/px)	Resample tiles to a fixed ground size	Blank (native pixels)
Drop tiles below vegetation %	Grid mode filter, keeping some empty tiles as hard negatives	Default

Exact copies. With tif or png and Bits = same as plot image, every tile holds exactly the values of its window in the plot image; georeferencing and compression do not change them. QA (tab 7) checks each tile pixel by pixel (IDENTICAL, or EXACT CONVERSION for another bit depth) and checks its georeference. Only jpeg, another bit depth or a fixed ground scale change the values.

9.3 How the rows are found

The rows are found in every plot image by row_detect.py (the PhenoFuse 4.0 row method, adapted to raw photos, where the rows are not always parallel to the image edge). Six steps: a greenness value per pixel, a threshold that separates plants from soil, a profile across the plot, the row spacing, the tilt, and a comb fitted to the rows. The figures show each step on OzBarley 2025 plot 1001 (a young crop, rows tilted 2.8° in the photo, 5 sown rows). The numbers are those the tool computed for this plot.

Step 1 · Greenness of every pixel (excess green)

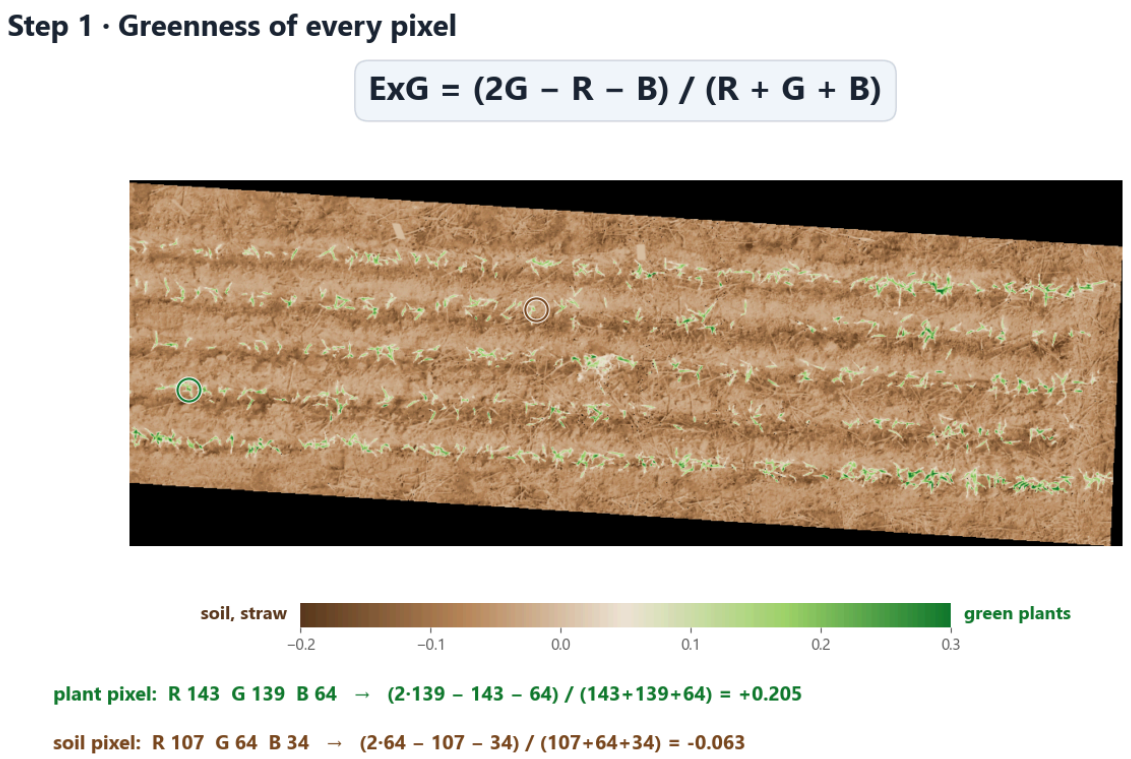


Figure 16. *The plot image turned into excess green: plants high (green), soil and straw low (brown). Two example pixels with their calculation.*

Every pixel gets **ExG = (2G − R − B) / (R + G + B)**. Dividing by R + G + B removes brightness, so a shaded leaf and a sunlit leaf score about the same. Green plants score high (about 0.1 to 0.3); soil and straw about 0 or below. Black pixels (outside a masked plot) are ignored.

Step 2 · Plant or soil: the Otsu threshold

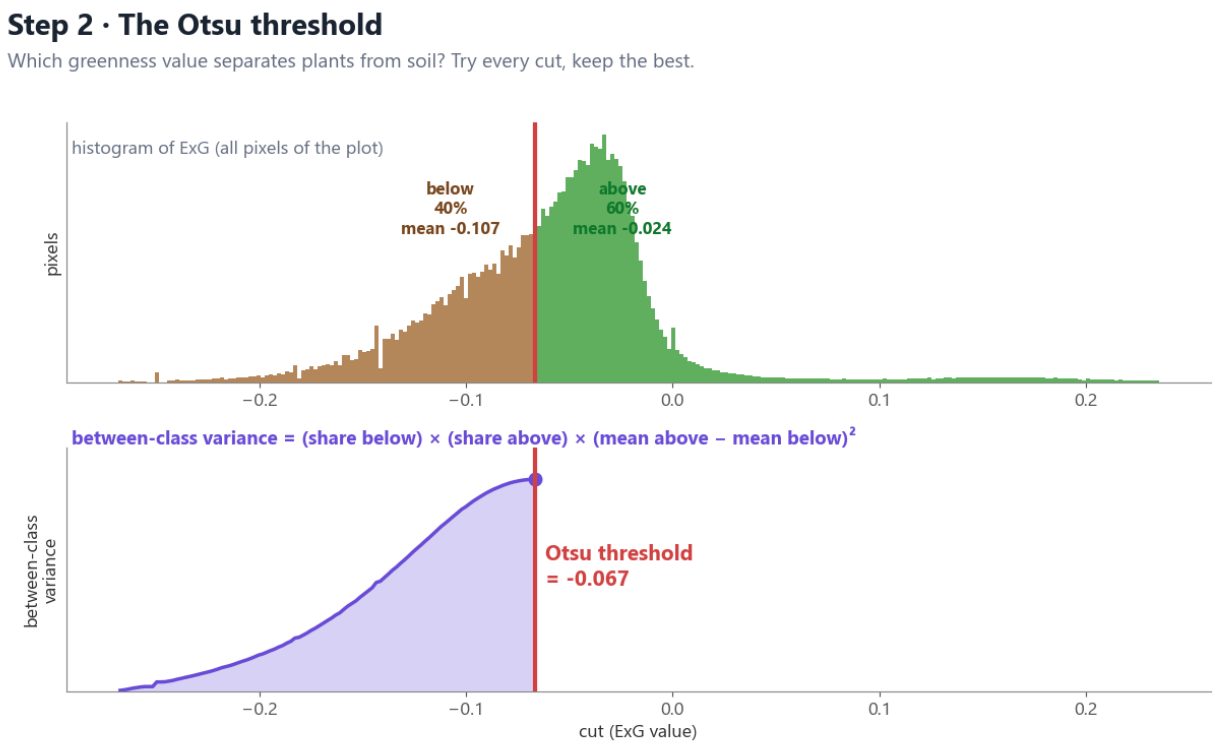


Figure 17. *Otsu's method on the ExG histogram of plot 1001: every cut is tried; the between-class variance (bottom) peaks at the threshold.*

Above which ExG value is a pixel a plant? **Otsu's method** answers this from the histogram alone, with no setting. It tries every possible cut. Each cut splits the pixels into two groups, below and above, and gets a score, the **between-class variance**:

between-class variance = (share below) × (share above) × (mean above − mean below)²

The score is large when both groups are big and their means are far apart. The cut with the largest score is the Otsu threshold: the cut that separates the histogram into the two most distinct groups. It adapts to every plot and every light condition.

Step 2b · When Otsu splits the soil (green fallback)

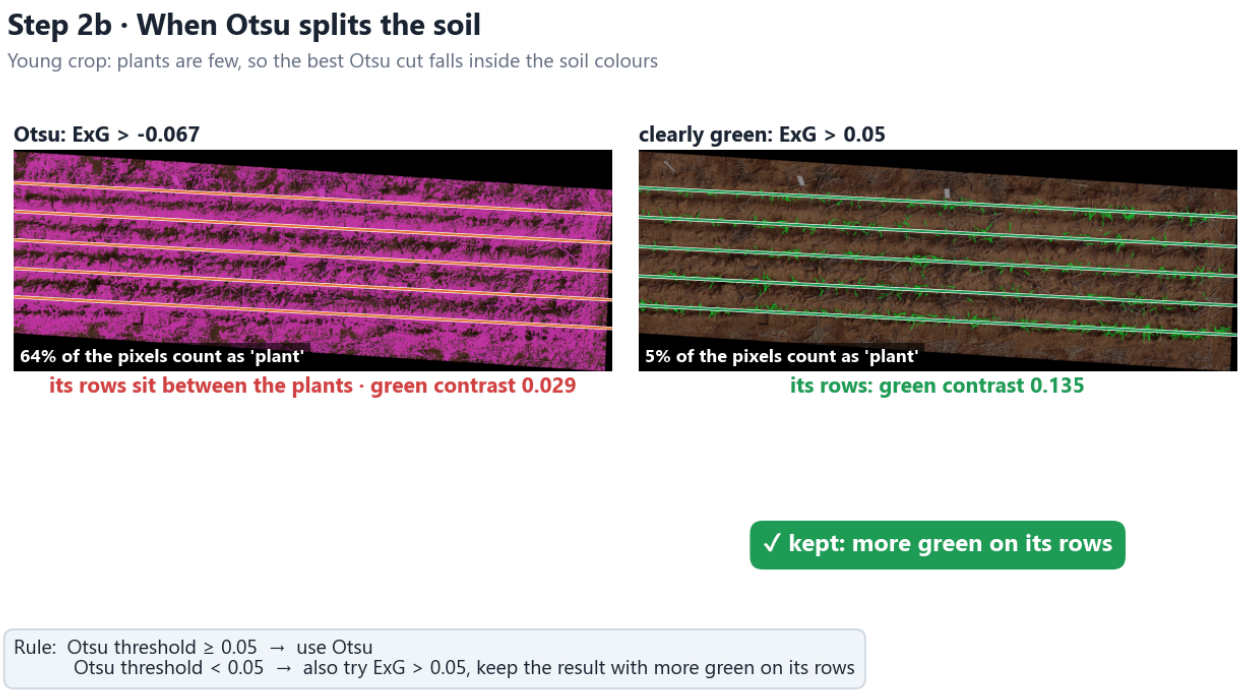


Figure 18. *Left: Otsu's cut (−0.067) falls inside the soil colours; 64% of the pixels count as plant and the rows found sit between the plants. Right: clearly green pixels (ExG > 0.05); its rows sit on the plants and are kept.*

Otsu always finds two groups, even when one of them is not plants. In a young crop the plants are a few percent of the pixels, so the best cut falls inside the soil colours: lighter soil and straw count as plants, and on ridged seedbeds the rows can then be found on the ridges, half a spacing off. So:

- **Otsu threshold ≥ 0.05:** the plants dominate the split; the Otsu result is used.
- **Otsu threshold < 0.05:** the rows are also found from the clearly green pixels only (ExG > 0.05). Both results are scored by how much more green sits on their rows than in the gaps (green contrast), and the higher one is kept. Plot 1001: Otsu 0.029, green 0.135, so the green result is used.

Step 3 · The row profile

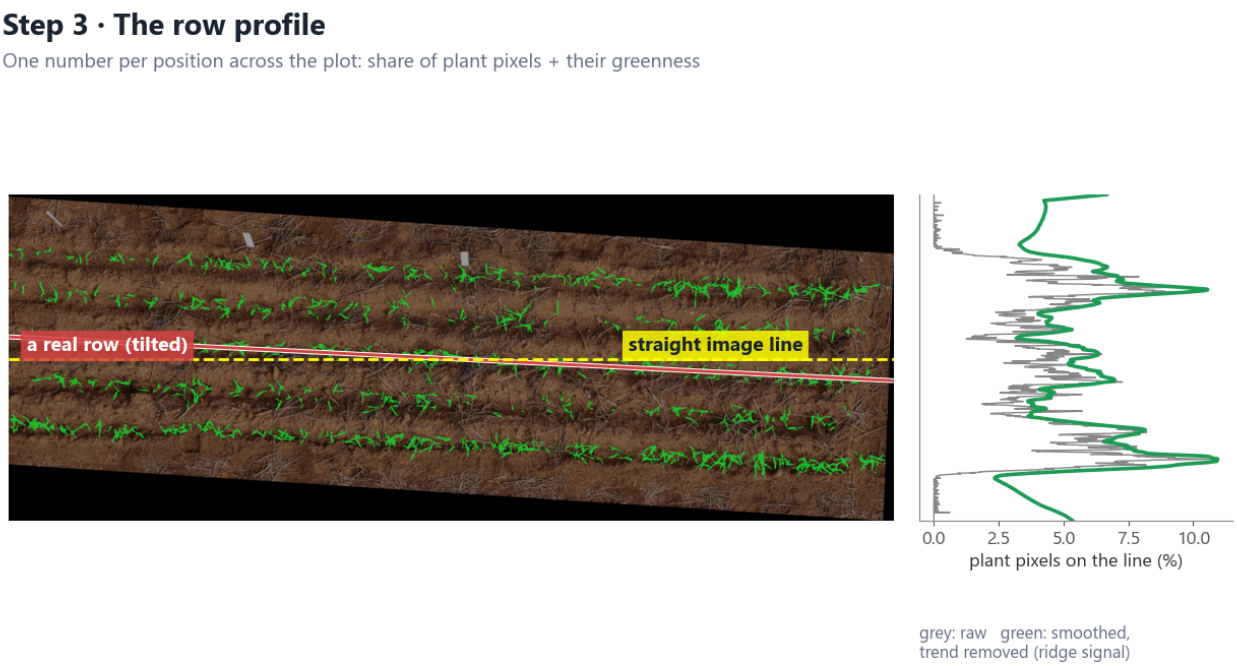


Figure 19. *Right: for each position across the plot, the share of plant pixels along that line (grey) and the smoothed ridge signal (green). The tilted rows smear the peaks of a straight image line.*

For each position across the plot one number is computed along a line in the row direction: the **share of plant pixels** on that line plus their **mean greenness** (the second helps at emergence, when seedlings are only a few pixels wide). Rows become peaks and the soil between them valleys. The profile is smoothed over 4 cm and the slow trend over 0.5 m is removed, so only the row pattern is left (the ridge signal). On a tilted plot a straight image line crosses several rows, so the spacing and the tilt are measured first.

Step 4 · Row spacing from short segments (Fourier)

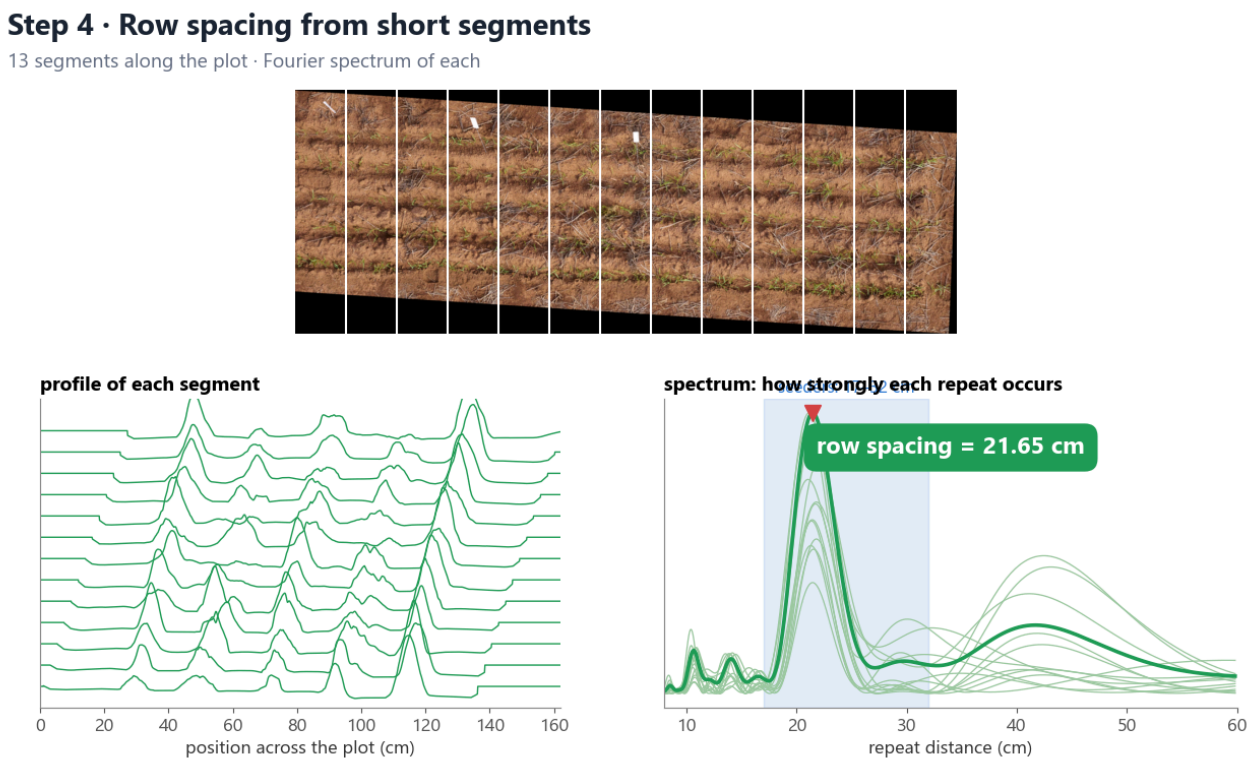


Figure 20. *The plot cut into 13 segments; the profile of each (left) and its spectrum (right). The sum of the spectra peaks at the row spacing.*

The plot is cut into 6 to 24 short segments along its length (13 here). Over a short segment even a tilted row stays sharp. Each segment's profile goes through a **Fourier transform**, which shows how strongly each repeat distance occurs (the position of the rows does not matter there, so the tilt cannot bias it). The spectra are added up and the strongest repeat between **17 and 32 cm** (the plot-seeder range; or exactly the row spacing if you enter one) is the row spacing. It is then refined on the whole, tilt-corrected plot: 21.65 cm here.

Step 5 · Tilt from the drift of the rows

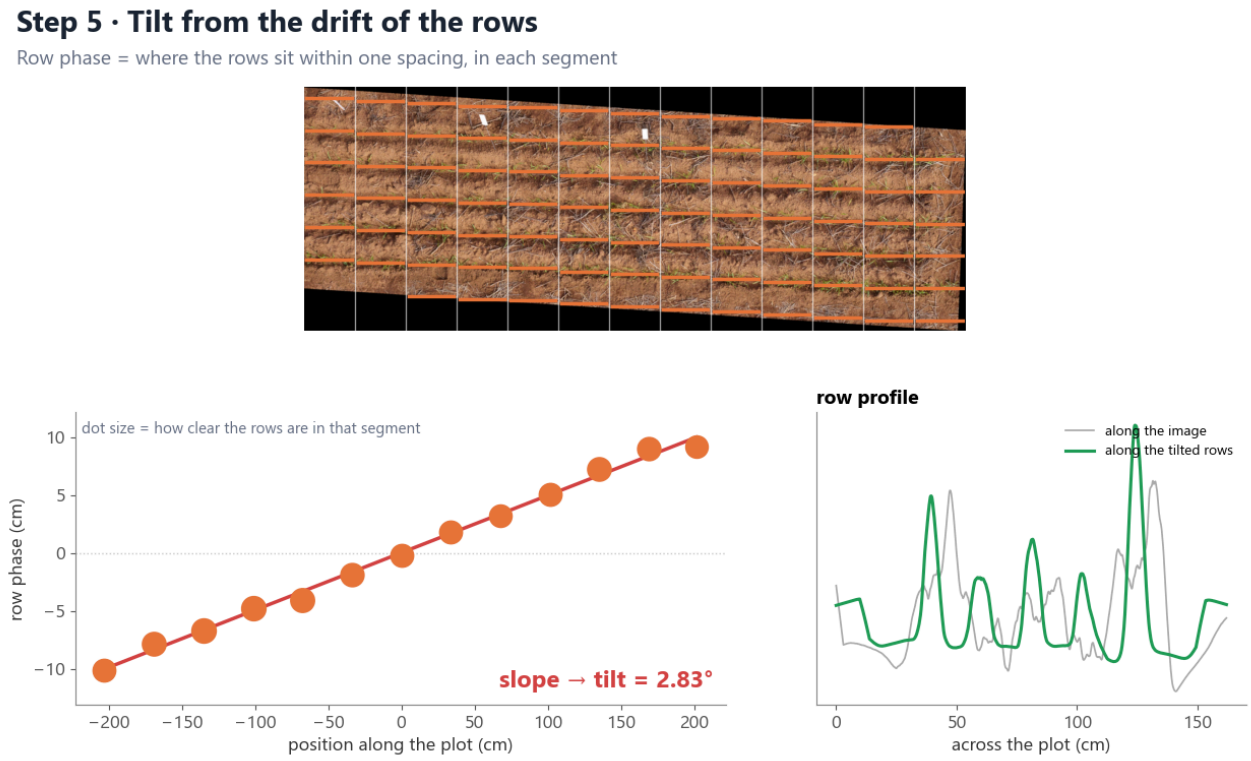


Figure 21. *Top: the row positions measured in each segment. Bottom left: their phase drifts steadily along the plot; the fitted line's slope is the tilt (2.83°). Bottom right: the profile before and after the tilt correction.*

In each segment the **phase** of the rows is measured: where the rows sit within one spacing (from the same Fourier component). If the rows ran exactly along the image, the phase would be the same in every segment; on a tilted plot it drifts steadily along the plot. A straight line is fitted through the phases (weighted by how clear the rows are in each segment, outliers dropped, 4 rounds); its slope is the **tilt** (a tilt above 8° is treated as no tilt). The profile is then measured along the tilted lines, and the row peaks become sharp.

Step 6 · The comb fit

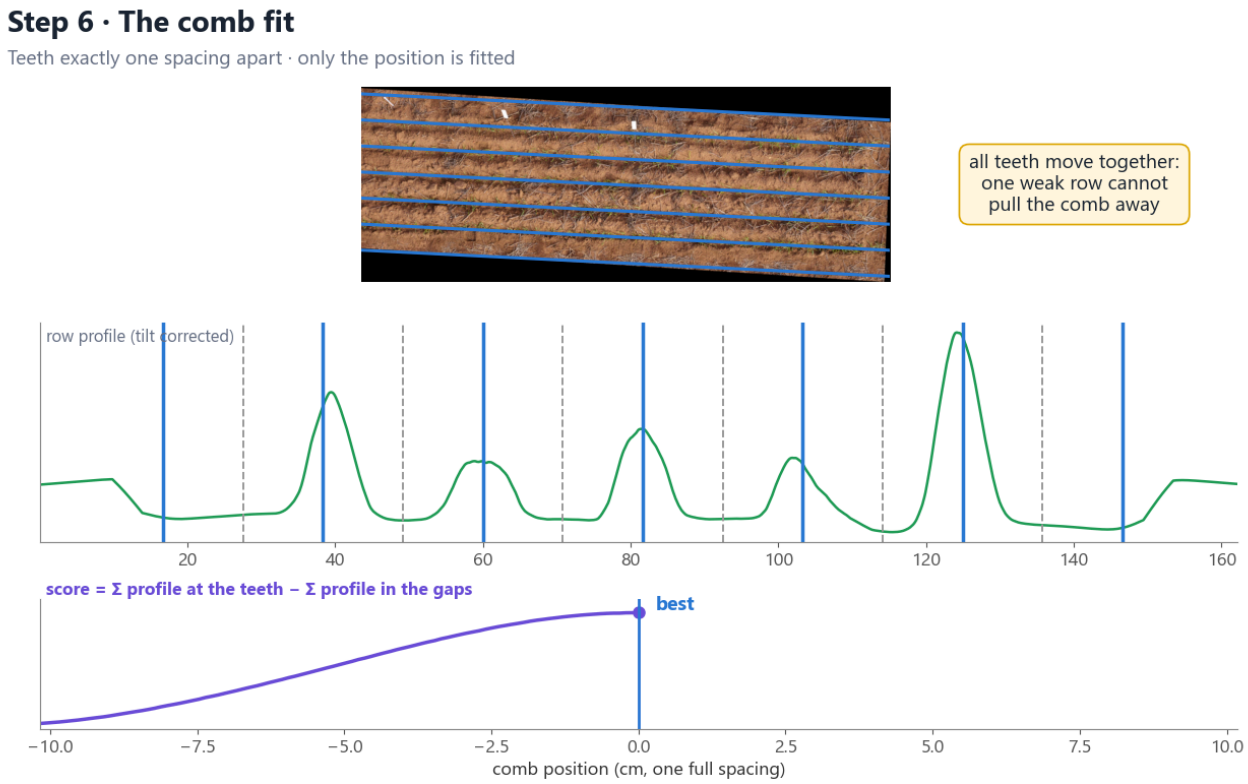


Figure 22. *The comb (blue teeth, one spacing apart) at its best position; dashed lines are the gaps between the teeth. Bottom: the score for every comb position across one spacing.*

A **comb** is a set of teeth exactly one row spacing apart, enough to fill the plot. The spacing is known, so only the comb's position is fitted: it slides across one full spacing, and every position gets a score:

score = Σ profile at the teeth − Σ profile halfway between the teeth

The score is high when the teeth sit on rows and the gaps on soil; the best position wins. Because all teeth move together, one weak, gappy or missing row cannot pull the comb away: the other rows hold it in place. That is why a comb is more reliable than picking the peaks one by one.

Which teeth are the plot's rows

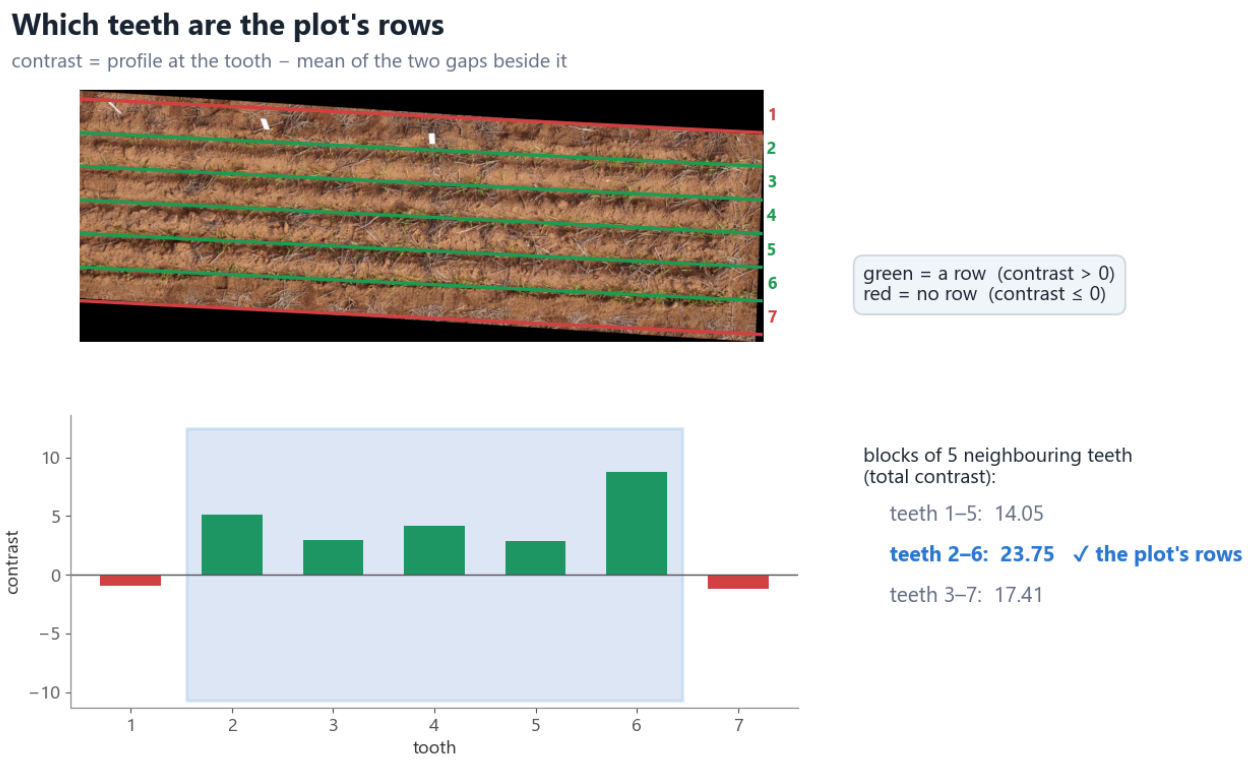


Figure 23. *Contrast of every tooth (green = a row, red = no row) and the block of 5 neighbouring teeth with the highest total contrast.*

Every tooth gets a **contrast**: the profile at the tooth minus the mean of the two gaps beside it. Positive = a real row (green), zero or negative = no row (red, e.g. bare ground at the edge of the crop). With **sown rows per plot** = 5, the block of 5 neighbouring teeth with the highest total contrast is the plot's rows (teeth 2-6 here); this keeps out rows of the neighbouring plots that the margin catches. With sown rows 0 the run of positive teeth is used, without the teeth cut by the crop edge.

Middle rows, band and tiles



Figure 24. *The middle 3 of the 5 rows (yellow), the band (half a spacing beyond the outer two, following the tilt), the 0.5 m end zones (red) and the 4 tiles.*

The **band** of the N middle rows runs from half a row spacing outside the first middle row to half a spacing outside the last, so a square tile spanning the band holds exactly N rows. It follows the tilt. The tiles are placed along it, inside the usable length (section 9.4), and each is an exact copy of its window in the plot image. rows_report.csv lists the rows, spacing and tilt of every plot, and _rows_qa\ has a picture of each.

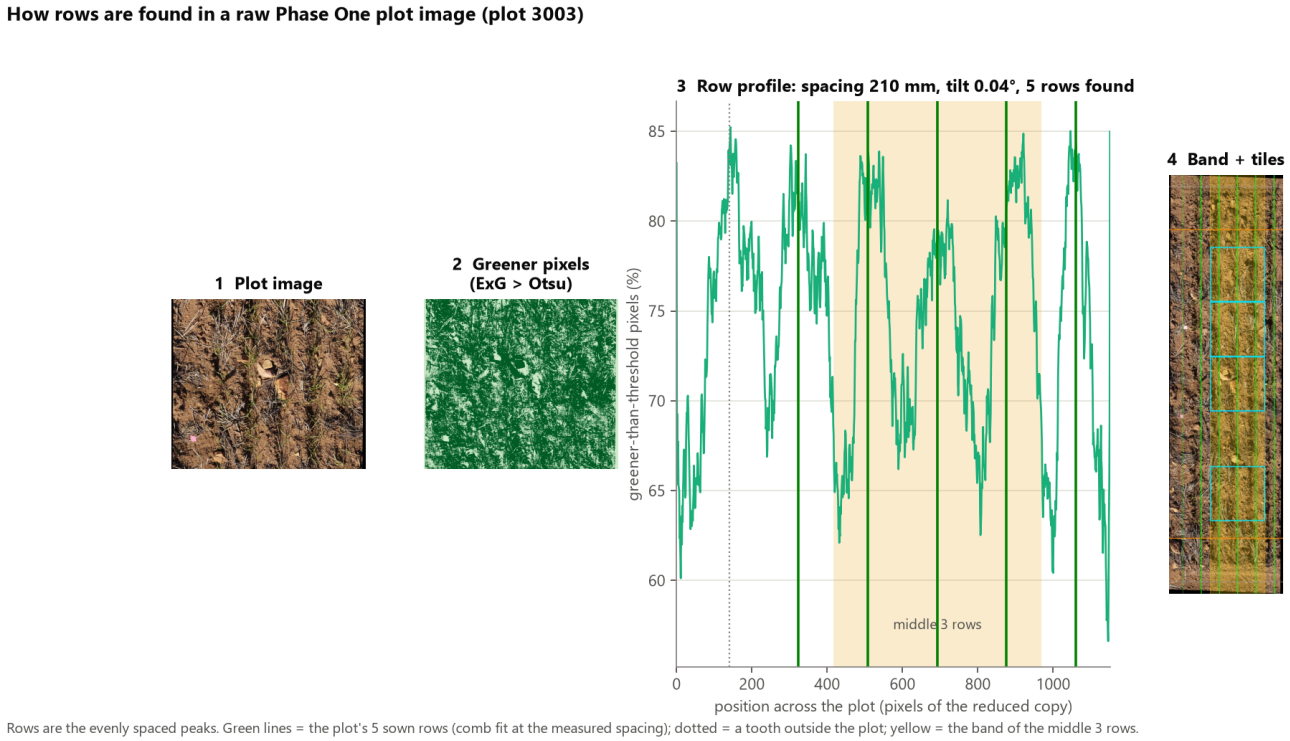


Figure 25. *Summary on another plot (York plot 3003): image, plant pixels, row profile with the fitted rows and the middle band, band and tiles.*

9.4 Where the tiles go along the plot

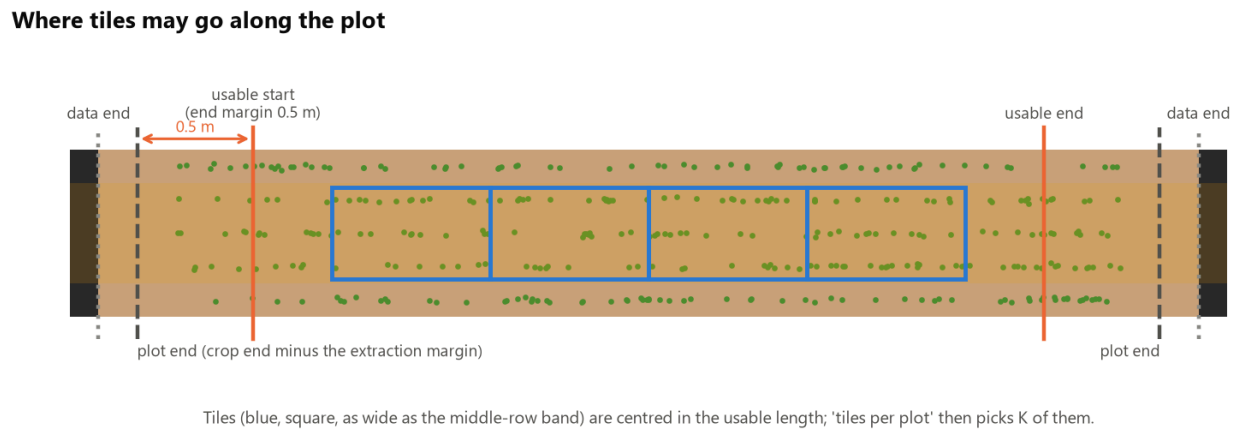


Figure 26. *Tiles stay inside the usable length and are centred in it.*

The plot ends are bare soil (and in a raw crop the black mask or the photo edge). Tiles there would teach a model that empty ground belongs to a plot. So the tiles stay inside the **usable length**: the plot ends (crop ends minus the extraction margin, read from manifest.csv), trimmed to the planted length where the green share clearly drops at the ends, minus the **end distance** you set. The tiles are centred in that length, then **Tiles per plot** picks how many are written.

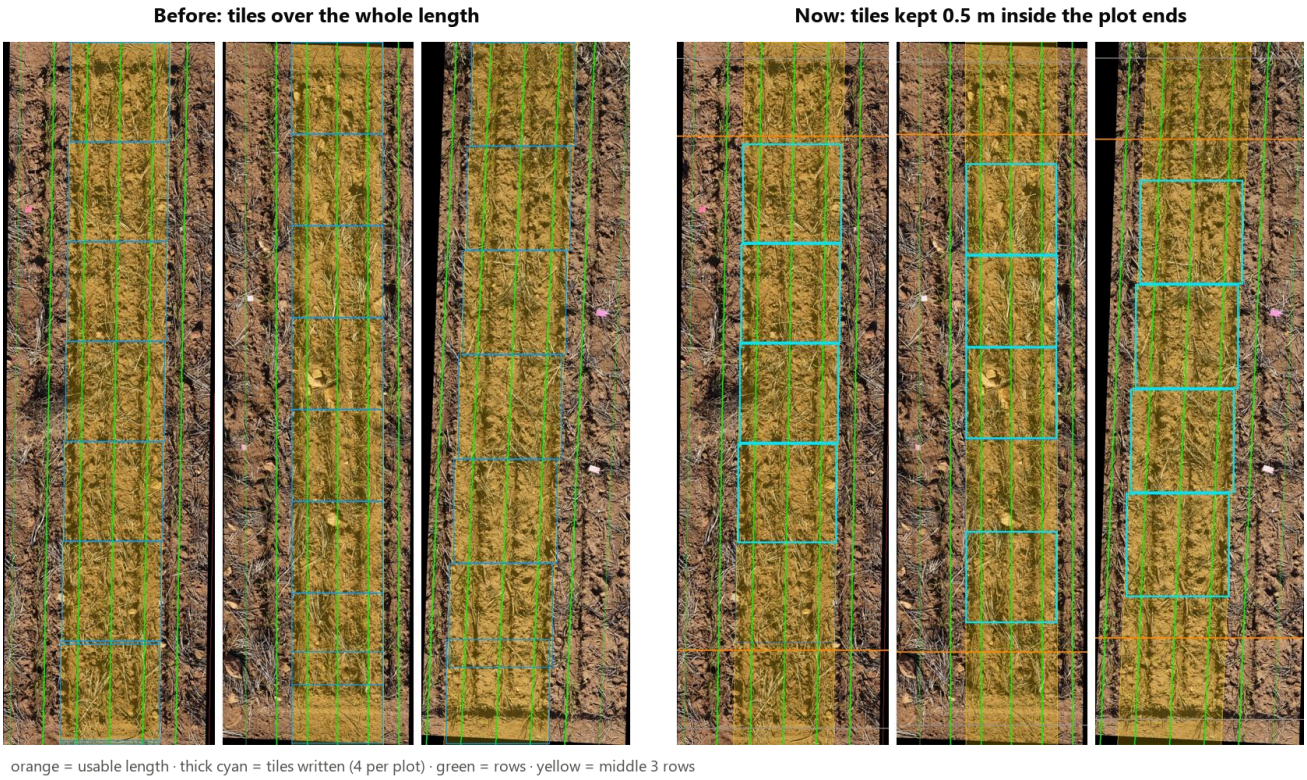


Figure 27. *Before the update tiles ran the whole plot length; now they stay 0.5 m inside the plot ends.*

9.5 What you get

- <plot>\<plot>_tile_c<i>_r0.png: the tiles, exact copies of the crop pixels (no resizing).

- tile_index.csv: per tile, the plot and the pixel window in the crop. Needed to map a detection back to its plot.
- rows_report.csv: per plot, rows found, spacing (px and mm), tilt, which rows were used, band, plot length, usable length, tiles that fit, tiles written.
- _rows_qa\<plot>_rows.jpg: green = rows, yellow = middle-row band, grey = plot ends, orange = usable limits, thick cyan = tiles written (numbered), thin blue = tiles not picked.



Figure 28. *Four middle-3-row tiles of plot 3003 (raw, original pixels).*

9.6 Other tools on this tab

- **Plain grid tiles:** untick middle rows. Fixed-size tiles with overlap over the seed-row band or the whole crop.
- **Rebuild one image per plot:** puts tiles (or labelled overlays with the same names) back together using tile_index.csv.

10. Tab 7 · QA, and are the images really original?

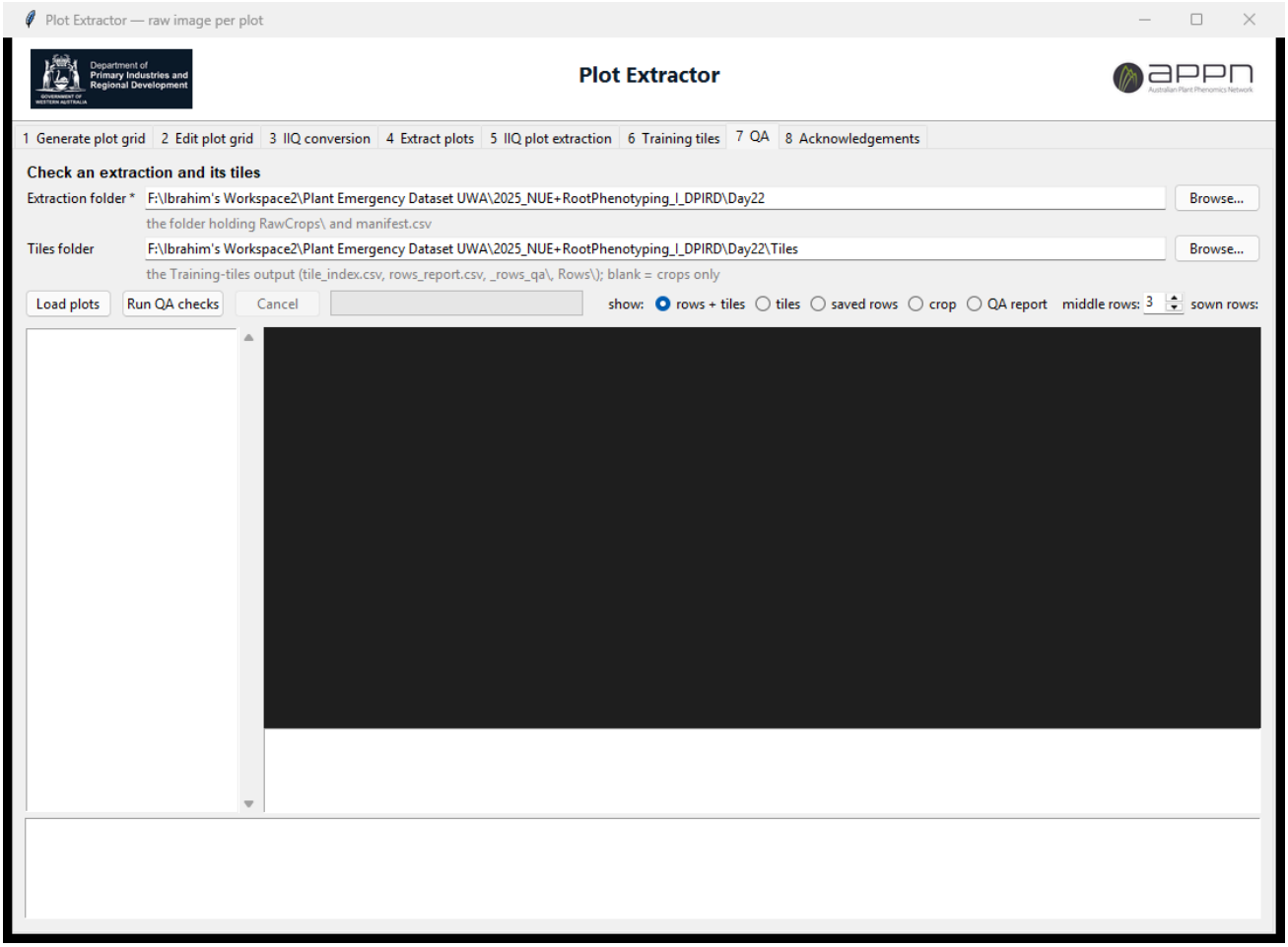


Figure 29. *Tab 7, QA.*

Tab 7 checks an extraction (and its tiles): every raw crop against its source - the JPEG box must be identical; an IIQ crop is compared with a fresh development of its box (IDENTICAL, or MATCH within 2/255 with noise reduction) - every tile against the crop, and the rows. It writes QA\QA_REPORT.txt with CSVs and shows per-plot pictures.

Tested on the York calibration trial (Phase One, 25 June 2025, 36 wheat plots):

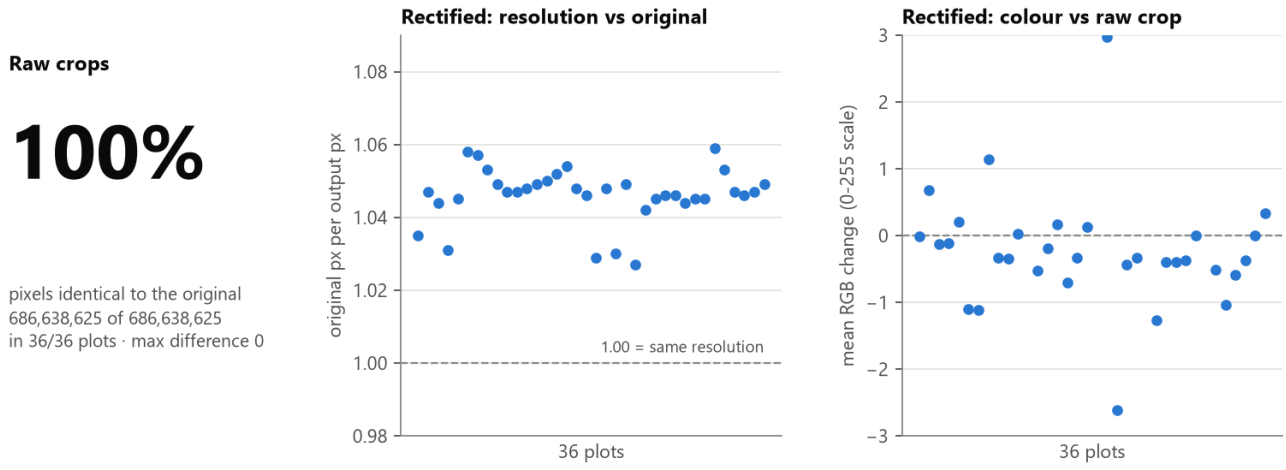


Figure 30. *Raw crops are identical to the original photos; rectified crops are resampled with unchanged colour.*

- **Raw (Masked to plot):** 36/36 plots, 686,638,625 pixels compared with the original JPEGs, every pixel identical. Colour profile copied byte for byte; lossless files.
- **Rectified:** resampled by design, about 1.05 original pixels per output pixel (~5% lower resolution); colour unchanged.
- **Tiles:** every tile is an exact copy of its window in the crop, so raw tiles are original pixels.

The check can be re-run on any extraction with fidelity_check.py (in the York output's fidelity\ folder).

11. Tab 8 · Acknowledgements

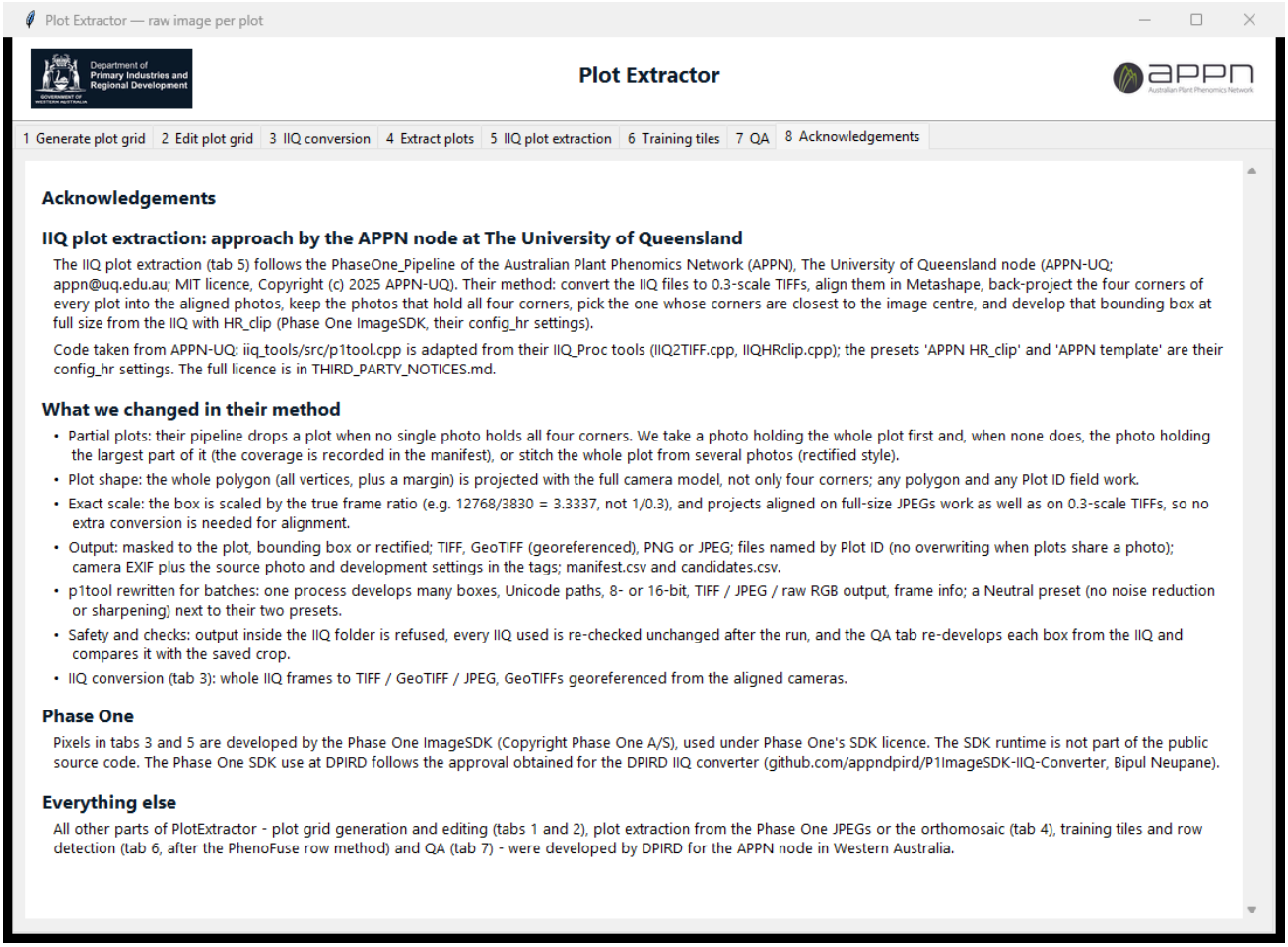


Figure 31. *Tab 8 lists the credits and our changes.*

The IIQ plot extraction (tab 5) follows the **PhaseOne_Pipeline of APPN, The University of Queensland** (appn@uq.edu.au, MIT licence): back-projecting the plot corners into the aligned photos and developing the plot box from the IIQ with HR_clip. Pixels in tabs 3 and 5 are developed by the **Phase One ImageSDK** (Copyright Phase One A/S). Tab 8 lists what we changed in their method: partial plots (largest part) and stitching instead of dropping plots, the whole polygon with a margin, the exact frame scale, masked/rectified GeoTIFF output named by Plot ID, a batch p1tool with a Neutral preset, protection and re-checks of the IIQ files, QA, and whole-frame conversion with georeferencing. Everything else in PlotExtractor was developed by DPIRD for the APPN node in Western Australia.

12. Tips and good practice

12.1 Test first

Run Limit = 3, open the crops and the tile QA pictures, then run everything.

12.2 Use a small margin

A plot grid is rarely exactly on the rows (at York it was about 9 cm off). A 0.10 m margin keeps the outer rows whole, which makes the row choice reliable, and does not add neighbouring rows when the inter-plot gap is ~40 cm.

12.3 Give the sown rows

With the number of sown rows set, all 36 York plots used rows 2-4 of 5 in both the raw and rectified sets. Without it, a row cut by the plot edge can shift which rows count as the middle ones.

12.4 Flight planning for whole plots in one photo

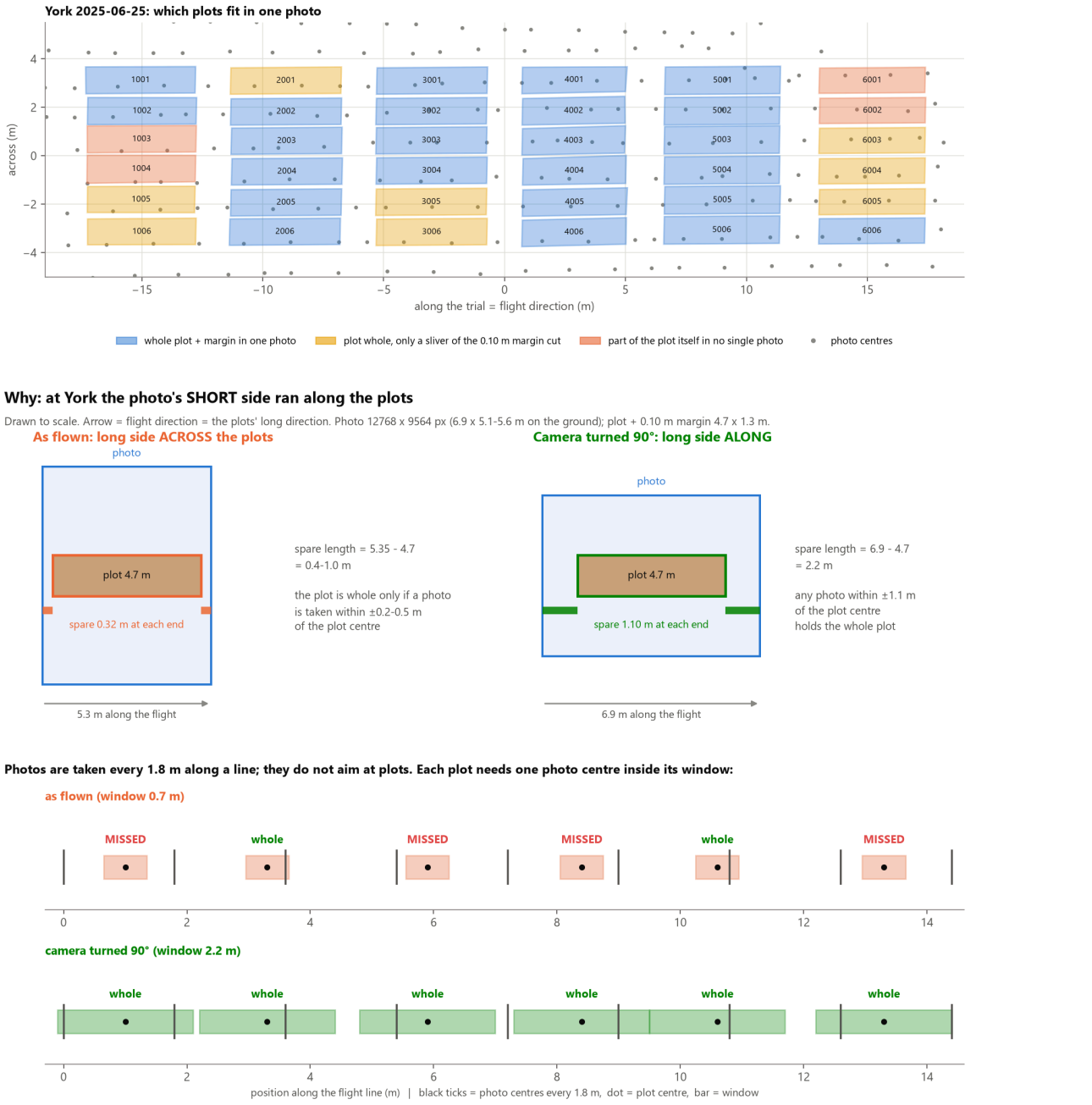


Figure 32. *Top: York plots coloured by fit. Middle: one photo and one plot drawn to scale, as flown and with the camera turned 90°. Bottom: each plot needs a photo centre inside its window; photos come every 1.8 m.*

At York (25 June 2025) the flight lines ran along the plots, but the photo's **short side** (9564 px, 5.1-5.6 m on the ground) lay along them and the long side (12768 px, 6.9 m) across. A plot plus its 0.10 m margin is 4.7 m long, so a photo was only 0.4-1.0 m longer than the plot: the **spare length**. The whole plot is in a photo only if that photo was taken within half the spare length (±0.2-0.5 m) of the plot centre: the plot's **window**.

The camera does not aim at plots: it takes a photo every **1.8 m** along the line (median of 762 photo pairs). A window narrower than 1.8 m catches a photo centre only by luck. Most plots were lucky on at least one flight line; 4 of 36 never were, so their raw images are cut at the photo edge.

Rule: spare length > distance between photos. Spare length = photo length along the plots − plot length (with margin). Then at least one photo centre always falls inside every plot's window, on every flight line.

- **Turn the camera 90°** (rotate it about the vertical axis) so the photo's long side runs along the plots: 6.9 − 4.7 = 2.2 m to spare, more than the 1.8 m between photos, so every plot is whole in at least one photo on every line.
- **Check a flight before processing:** open one photo. The plots must run along its **long** side. (Pointing the camera straight down, −90° gimbal pitch, is a different setting and does not change this.)
- **Or take photos more often:** the distance between photos must be below the spare length (here below 0.4-1.0 m, usually not practical).
- **Or fly 2-3 m higher** (slightly lower resolution).
- **Start and end flight lines a few metres past the trial** so the end plots get the full number of photos.

12.5 Check the QA pictures

Open a handful of _rows_qa pictures: the green lines should sit on the rows, the yellow band on the middle rows and every cyan tile between the orange lines.

13. Troubleshooting

Message or symptom	Cause and fix
No valid Metashape license	Licence server not reachable: connect to the network/VPN.
plots are ~X km away from the project centre	Wrong CRS or wrong .psx. Check the shapefile's .prj or set Plot CRS.
raw file missing	Set Raw images folder to where the photos are (the JPEG folder).
Chunk 'X' not found	The chunk saved from another project: pick the chunk again (or (active)) after choosing the .psx.
Phase One SDK status not OK (tabs 3, 5)	Leave the SDK folder blank to use the bundled SDK, or point it at an SDK with ImageSdkCpp.dll and SensorProfiles.
No .IIQ file matches the cameras (tab 5)	The IIQ folder must hold the flight's raw files with the same names as the photos in the project (T0_Raw).
frame left without georeference (tab 3)	The camera is not aligned in the chosen chunk, or the project/chunk is not the one the photos were aligned in.
raw_covered_pct below 100	No single photo holds the whole plot. Use the rectified image for that plot, or see 12.4.
no middle-row tiles (only K rows found)	Rows not visible enough or too few rows. Check the QA picture; set sown rows; use a larger extraction margin.
usable length shorter than one tile	Plot too short for the band width plus end distance: reduce 'keep tiles … m from the plot ends'.
only K tiles fit, N asked	The plot holds fewer tiles than asked; all that fit are written.
Middle rows look off by one row	Set the sown rows per plot and re-extract with a 0.10 m margin.

Appendix: files

File	Contents
manifest.csv	plot_id, plot_area_m2, margin_m, raw_image, raw_file, crop_w/h, corners_in_frame, raw_covered_pct/m2, cand_frames, best_frame, best_cover_pct/m2, frames_used, seam_y_pct, ortho_*, source_path, raw_style, src_x0, src_y0, src_x1, src_y1, note
candidates.csv	plot_id, rank, camera, cover_pct, cover_m2, corners_in_frame, aniso, mm_per_px, used
tile_index.csv	tile, rel_path, plot_stem, col, row, x0, y0, w, h, out_px, src_mm_px, tile_mm_px, ground_mm, band_lo, band_hi, across_axis, src_w, src_h, veg_pct, src
rows_report.csv	plot_stem, src, ok, reason, rows_found, rows_present, pitch_px, pitch_mm, angle_deg, middle_rows, band_lo, band_hi, plot_len_m, end_trim, usable_lo, usable_hi, usable_len_m, tiles_fit, tiles_in_band, tiles_drawn